

Persistenza dell'effetto Host nelle Olimpiadi controllando variabili macroeconomiche



Daniele D'Ugo - 0001186581

Simona Paparesta - 0001019811

Tommaso Compagnucci - 0001176284

Human Data Science – A.A 2025-2026

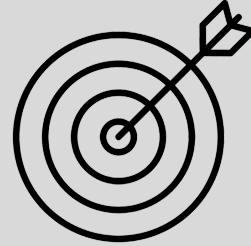
Obiettivo dello Studio



Giornali e opinioni comuni associano spesso l'ospitare le Olimpiadi a un vantaggio competitivo (home advantage)



Tuttavia, tali considerazioni non tengono conto di altri fattori, storici e macroeconomici



Obiettivo del nostro studio, è quello di verificare l'influenza dell'home advantage quando si tengono in considerazione anche tali variabili



Dati e Metodologie

Dati Olimpici

- 120 Years of Olympic History: Athletes and Result (1896 - 2016)
- Wikipedia (2020 - 2024)

Dati Macroeconomici

- World Bank Open Data
- Maddison Project

Metodologie

- Analisi Within Country e Sign Test
- Correlazione di Spearman
- Bayesian Changepoint Detection
- Regressione Multivariata

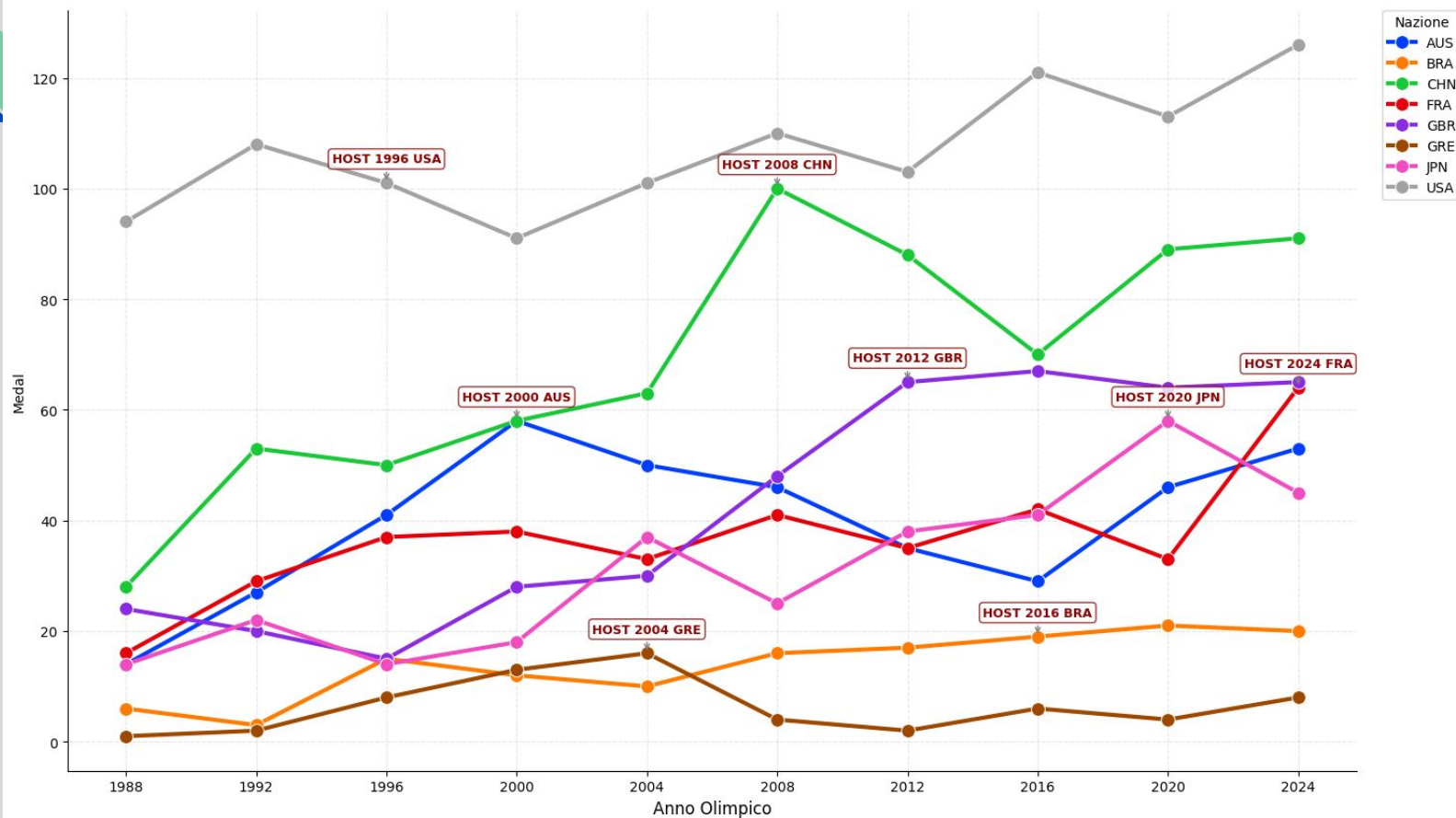


Analisi Within Country e Sign Test

- Confronto host vs se stesso (1964 - 2024)
- In 12 delle 13 edizioni Olimpiche analizzate, l'host ha avuto un incremento nella percentuale di medaglie vinte
- Il Test dei Segni ha poi riportato un p-value pari a 0.0034

JPN : =+2.2%	MEX : =+1.3%	GER : =+9.3%
CAN : =+0.1%	KOR : =+2.3%	ESP : =+1.5%
USA : =-1.3%	AUS : =+2.7%	GRE : =+1.3%
CHN : =+3.5%	GBR : =+3.0%	BRA : =+0.9%
	FRA : =+2.9%	

Confronto Storico dei Paesi Ospitanti





Correlazione di Spearman

Correlazione PIL -> Medaglie

- Media geometrica del PIL nei 4 anni precedenti
- I valori della correlazione, per ogni edizione, oscillano tra 0.64 e 0.69
- Paesi ricchi hanno più possibilità di investire nello sport

Correlazione Popolazione -> Medaglie

- Media geometrica della popolazione nei 4 anni precedenti
- I valori della correlazione, per ogni edizione, oscillano 0.43 e 0.54
- Paesi popolosi hanno a disposizione un bacino di talenti più ampio



Bayesian Changepoint Detection

- Utilizzato per trovare una svolta strutturale nel regime (changepoints)
- Ancora di ± 20 anni per eliminare importanti rumori storici
- Due modelli indipendenti per due serie storiche diverse

MEDAGLIERE

variabile discreta non negativa
Poisson-Gamma

GDP

variabile continua
framework Gaussiano



Il lead effect e il lead-lag

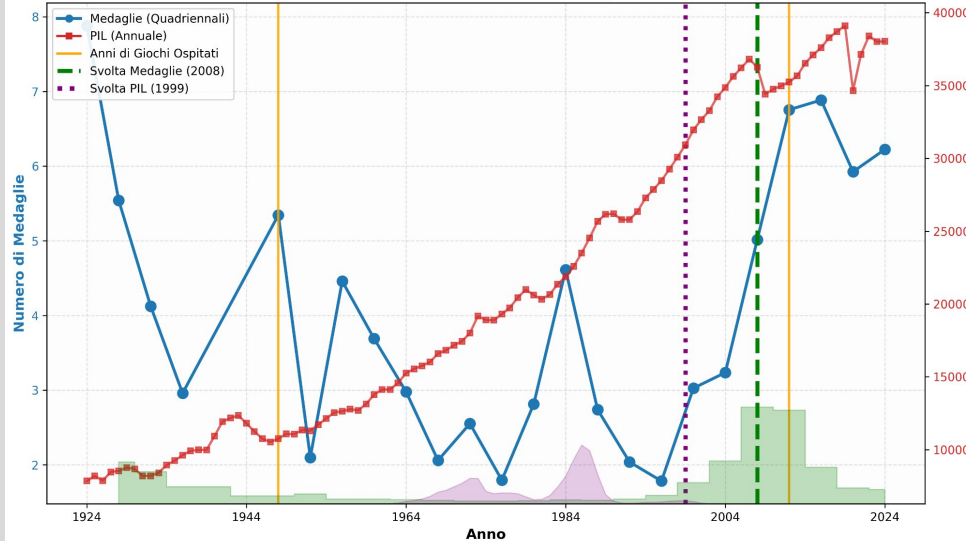
La crescita del **PIL** fornisce la copertura finanziaria per investimenti sistemici.

Il **rendimento atletico** si materializza con un ritardo variabile, consolidandosi nelle edizioni post-evento casalingo.

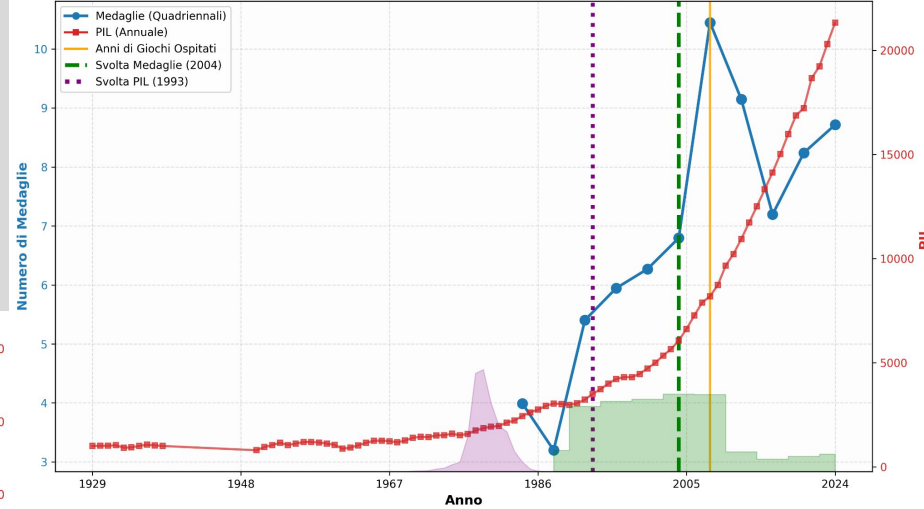
Distanza temporale tra MAP
sportivo ed economico:
 $\Delta T = T_{\text{medaglie}} - T_{\text{PIL}}$

Casi di conferma principali

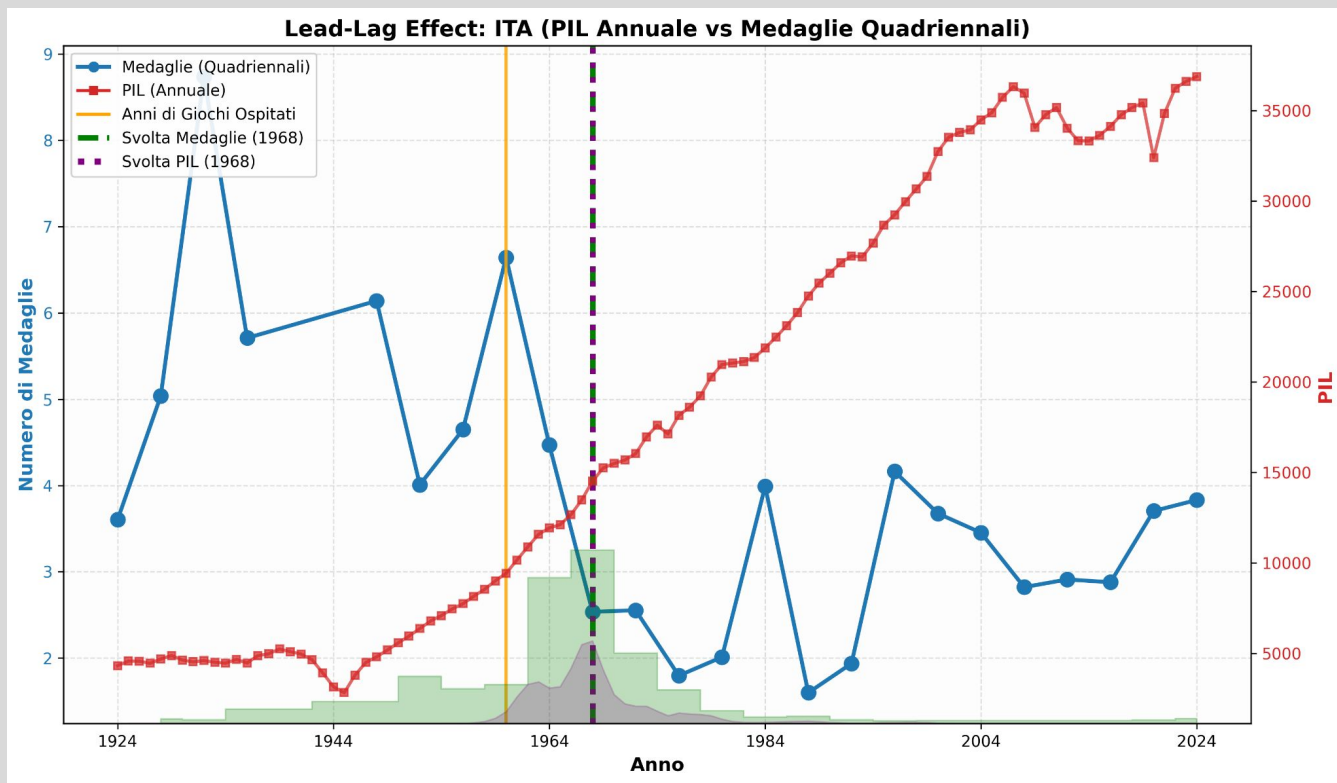
Lead-Lag Effect: GBR (PIL Annuale vs Medaglie Quadriennali)



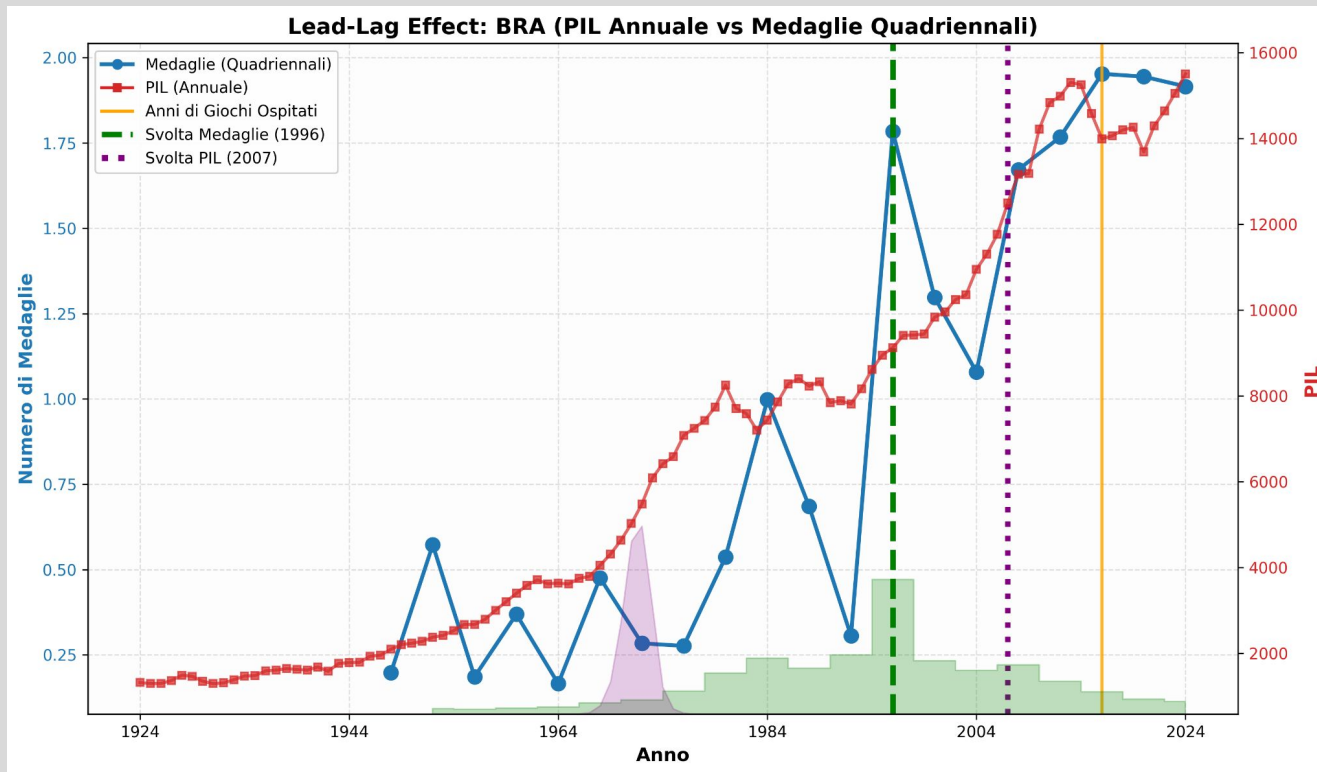
Lead-Lag Effect: CHN (PIL Annuale vs Medaglie Quadriennali)



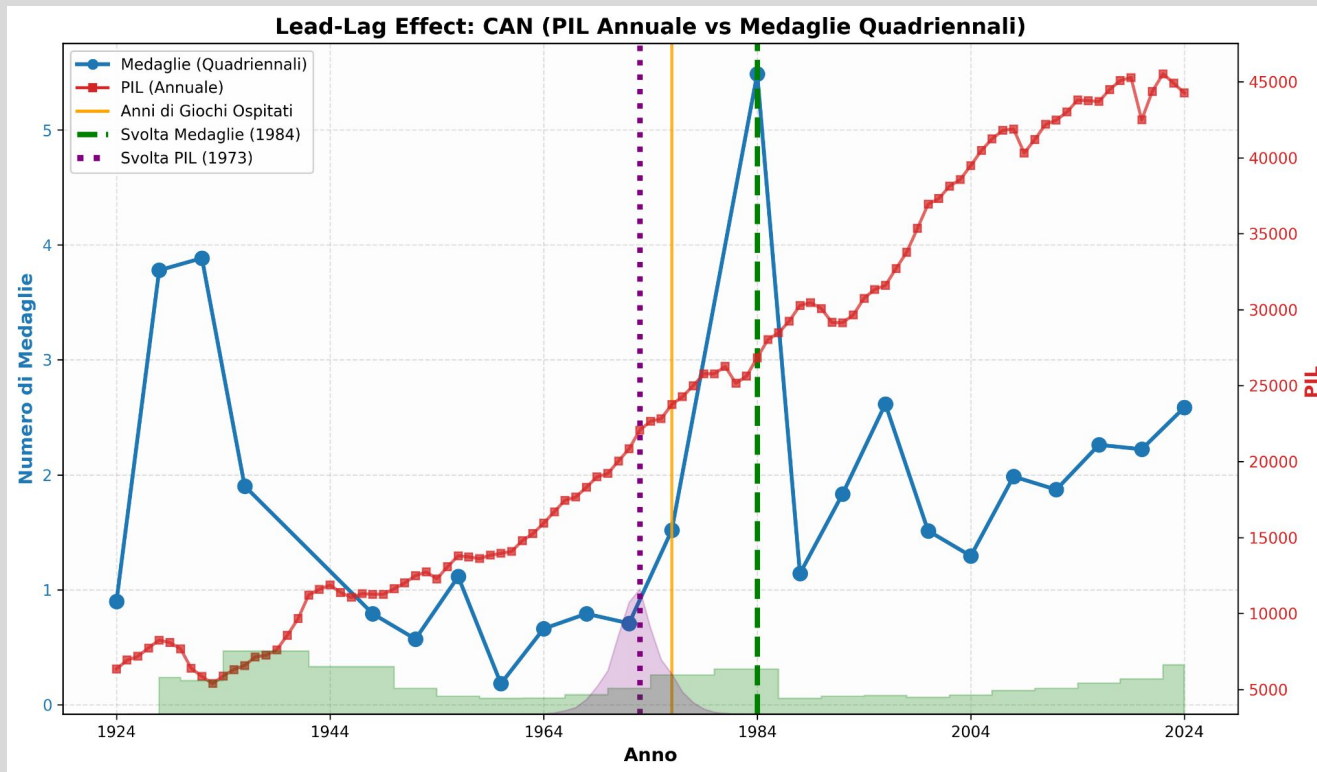
Il decoupling dell'Italia



L'inversione del Brasile



Il Lead-lag del Canada



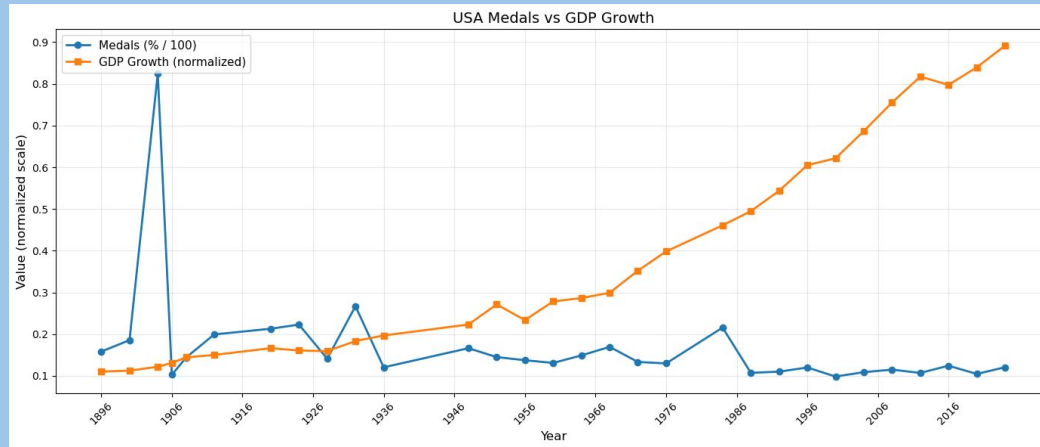
Dati e limitazioni



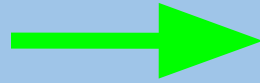
- **Non stazionari**
(ADF test)



~~**Granger
Causality**~~



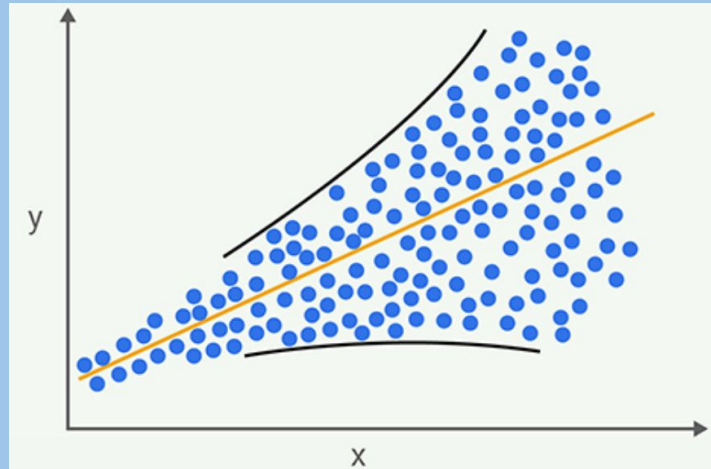
- Eteroschedasticità



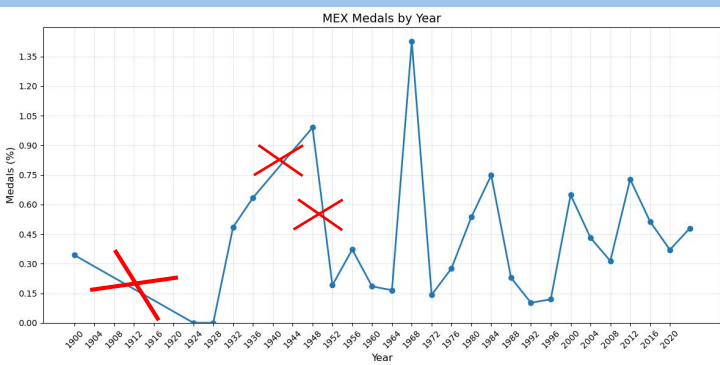
regressione con

HCO

(Heteroskedasticity-consistent
standard errors)



Serie interrotte



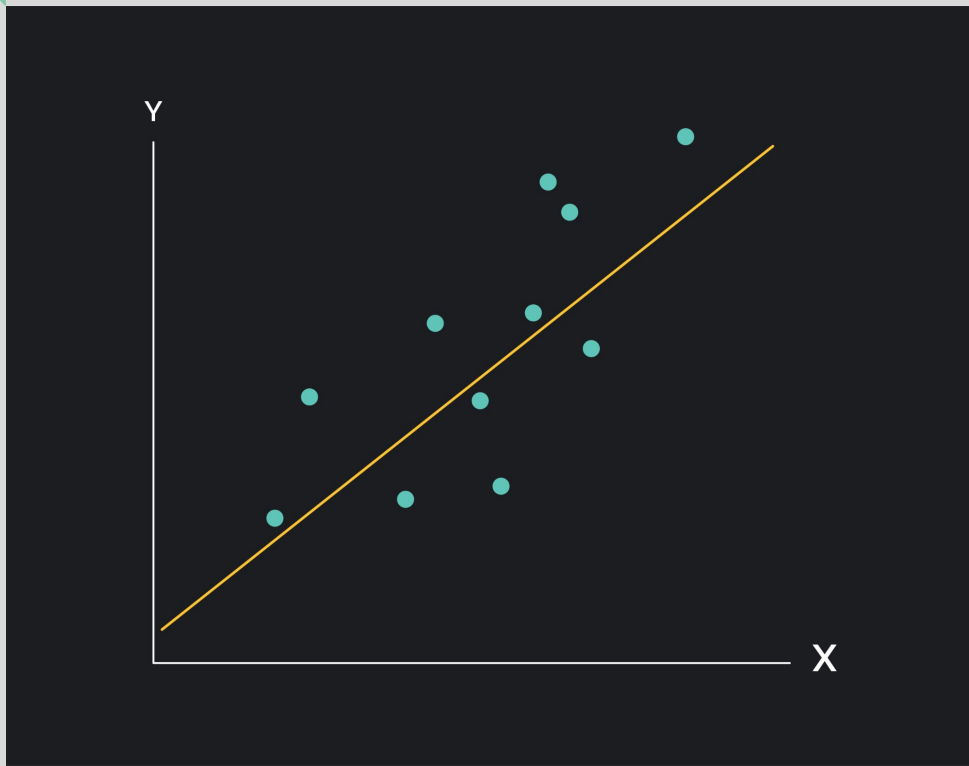
Guerre



**Paesi
vecchi...
e nuovi**



Regressione Multivariata



VARIABILI (HOST)

OG <year>



PRE <year>



POST <year>

VARIABILI (MACRO)



POPolazione

GDP_log
(PIL p.c.
pro capite)



COMMunist
Bloc

VARIABILI



AM

(Average Medals
History,
per Paese)

YEAR

<year>



2024

0 0 0 1

2020

0 0 1 0

2016

0 1 0 0

2012

1 0 0 0

1980



URS 195

USA 0

1984



URS 0

USA 173

BOYCOTT <year>

	Csurilla': ZINB	ZINB (YEAR)		OLS Top 40		
Variables	All	All	Top 40	YEAR	AM+YEAR	AM
R-squared	-	-	-	0.628	0.798	0.795
Adj. R-squared	-	-	-	(0.233)	(0.706)	(0.704)
	-	-	-	0.584	0.773	0.776
	-	-	-	(0.151)	(0.674)	(0.679)
Pseudo R-squared	-	0.1580	0.1400	-	-	-
	-	(0.01514)	(0.01953)	-	-	-
Log-Likelihood	-	-2902.7	-1096.3	-566.04	-470.41	-472.70
	-	(-2857.8)	(-1249.8)	(-679.32)	(-528.94)	(-530.29)
OG1996	0.021 (0.544***)	0.526*** (3.171***)	0.538*** (1.802***)	6.403*** (10.031***)	0.354 (-0.575)	0.656** (-0.255)
OG2000	0.361*** (0.476***)	1.706*** (2.496***)	1.129*** (1.154***)	3.850*** (4.294***)	4.201*** (4.076***)	4.377*** (4.211***)
OG2004	0.384 (0.273)	1.295*** (1.160***)	0.321*** (-0.186)	0.262 (-0.354)	0.160 (-0.372)	0.183* (-0.419***)
OG2008	0.101 (0.459**)	-0.114 (3.002***)	0.206 (1.673***)	4.100*** (8.492**)	5.186*** (7.897**)	5.128*** (7.828**)
OG2012	0.577*** (0.759***)	1.408*** (2.587***)	0.915*** (1.264***)	3.327*** (4.849***)	2.282*** (2.325**)	2.182*** (2.180**)
OG2016	0.362 (0.310)	0.255 (1.289**)	-0.489*** (-0.039)	-2.027*** (-0.078)	-0.403* (0.956***)	-0.500** (0.889**)
OG2020	0.114 (0.496**)	0.345*** (1.629**)	0.406*** (0.951**)	1.217*** (3.297**)	2.109*** (3.048**)	2.025*** (2.987**)
OG2024	—	0.834*** (1.755***)	0.819*** (1.079***)	2.686*** (4.048**)	1.913*** (2.031***)	1.867*** (2.056**)
PRE1996	0.181 (0.244)	1.507*** (2.269**)	0.895*** (0.901**)	2.463*** (2.897**)	2.765*** (2.628**)	3.096*** (2.893**)
PRE2000	0.446 (0.347)	1.063*** (1.000**)	0.112 (-0.340**)	0.087 (-0.571*)	-0.255 (-0.900**)	-0.109 (-0.765**)
PRE2004	-0.454* (-0.295)	-0.280 (2.532**)	-0.172 (1.183**)	0.499 (4.717**)	1.744*** (4.467**)	1.739*** (4.421**)
PRE2008	0.068 (0.205)	0.942*** (2.267**)	0.552*** (0.940**)	1.505*** (3.059**)	0.456** (0.466**)	0.428*** (0.407**)
PRE2012	0.387 (0.331)	0.164 (1.245**)	-0.542*** (-0.076)	-2.172*** (-0.141)	-0.511** (0.901**)	-0.632*** (0.738**)
PRE2016	-0.100 (0.306)	0.501*** (2.059**)	0.102 (0.730**)	-0.006 (2.183**)	0.963*** (1.932**)	0.902*** (1.872**)
PRE2020	-0.169 (0.002)	0.164 (1.064**)	0.109 (0.388*)	-0.407 (0.983*)	-1.137*** (-1.019**)	-1.233*** (-1.071**)
POST2000	-0.189 (0.375*)	0.298* (2.947**)	0.330*** (1.604**)	4.244*** (7.862**)	-1.573*** (-2.432**)	-1.427*** (-2.242**)
POST2004	0.233 (0.301)	1.624*** (2.300**)	0.998*** (0.952**)	2.853*** (3.314**)	3.318*** (3.275**)	3.367*** (3.228**)
POST2008	-0.886* (-0.953*)	-0.246* (-0.219)	-1.085*** (-1.544**)	-0.964*** (-1.539**)	-1.117*** (-1.640**)	-1.153*** (-1.712**)
POST2012	-0.139 (0.323)	-0.357* (2.890**)	0.009 (1.566**)	2.621*** (7.240**)	3.565*** (6.275**)	3.449*** (6.122**)
POST2016	0.778*** (0.975***)	1.404*** (2.550**)	0.880*** (1.220**)	3.404*** (4.855**)	2.351*** (2.348**)	2.276*** (2.301**)
POST2020	0.464*** (0.375*)	-0.100 (0.612**)	-0.416*** (-0.064)	-1.945*** (-0.129)	-0.422* (0.914**)	-0.546*** (0.845**)
POST2024	—	0.129 (1.403**)	0.218* (0.727**)	0.217 (2.228**)	0.952*** (1.842**)	0.916*** (1.858**)
ln_GDP	0.416***	0.898***	0.420***	0.848***	—	0.202***
ln_Population	0.178***	0.612***	0.426***	1.102***	—	0.620***
Is_Communist	0.696***	1.335***	0.438***	0.805***	—	0.671***
AM (History)	0.596*** (0.716***)	—	—	—	0.463*** (0.613***)	0.466*** (0.6132***)
Constant	-9.016*** (-1.911***)	-12.979*** (2.406***)	-5.849*** (3.079***)	-17.792*** (2.083***)	-7.934*** (0.700***)	(0.466***) (0.6132***)
Inflate Const	18.51*** (27.85***)	-14.745*** (-12.564***)	-21.808*** (-19.298***)	—	—	—
Alpha	—	1.179*** (3.781***)	0.218*** (0.652***)	—	—	—

OLS:

- problema: molti 0 → top 40
- continui → medaglie %
- multicollinearità

	VIF
AM	2.25
GDP	45.82
POP	51.02
COMM	1.52
OG	1.03 - 1.16
PRE	1.03 - 1.06
POST	1.03 - 1.15
YEAR	2.10 - 2.17

	Y00	Y04	Y08	Y12
00	1	0	0	0
04	0	1	0	0
08	0	0	1	0
12	0	0	0	1

ZINB:

- accetta 0 → top 40 / all
- discreti → medaglie (tot)
- multicollinearità

OG1996		0.021 (0.544***)
OG2000		0.361** (0.476***)
OG2004		0.384 (0.273)
OG2008		0.101 (0.459**)
OG2012		0.577*** (0.759***)
OG2016		0.362 (0.310)
OG2020		0.114 (0.496**)

PRE1996		0.181 (0.244)
PRE2000		0.446 (0.347)
PRE2004		-0.454* (-0.295)
PRE2008		0.068 (0.205)
PRE2012		0.387 (0.331)
PRE2016		-0.100 (0.306)
PRE2020		-0.169 (0.002)

POST2000		-0.189 (0.375*)
POST2004		0.233 (0.301)
POST2008		-0.886* (-0.953*)
POST2012		-0.139 (0.323)
POST2016		0.778*** (0.975***)
POST2020		0.464** (0.375*)

Csurilla (ZINB):

coeff. **con** MACRO
(coeff. **senza**
MACRO)

*p-value < **0.10**

p-value < **0.05

***p-value < **0.01**

ln_GDP		0.416***
ln_Population		0.178***
Is_Communist		0.696***
AM (History)		0.596*** (0.716***)

OLS

(AM, Top40):

ln_GDP 	0.202***
ln_Population 	0.620***
Is_Communist 	0.671***
AM (History) 	0.466***
	(0.6132***)

OG1996 	0.354 (-0.575)
OG2000 	4.201*** (4.076***)
OG2004 	0.160 (-0.372)
OG2008 	5.186*** (7.897***)
OG2012 	2.282*** (2.325***)
OG2016 	-0.403* (0.956***)
OG2020 	2.109*** (3.048***)
OG2024 	1.913*** (2.031***)

PRE1996 	2.765*** (2.628***)
PRE2000 	-0.255 (-0.900***)
PRE2004 	1.744*** (4.467***)
PRE2008 	0.456** (0.466**)
PRE2012 	-0.511** (0.901***)
PRE2016 	0.963*** (1.932***)
PRE2020 	-1.137*** (-1.019***)

POST2000 	-1.573*** (-2.432***)
POST2004 	3.318*** (3.275***)
POST2008 	-1.117*** (-1.640***)
POST2012 	3.565*** (6.275***)
POST2016 	2.351*** (2.348***)
POST2020 	-0.422* (0.914***)
POST2024 	0.952*** (1.842***)

coeff. **con** MACRO
(coef. **senza**
MACRO)

*p-value < **0.10**
p-value < **0.05
***p-value < **0.01**

OLS

(AM, Top40):

ln_GDP 	0.202***
ln_Population 	0.620***
Is_Communist 	0.671***
AM (History) 	0.466***
	(0.6132***)

OG1996 	0.354 (-0.575)
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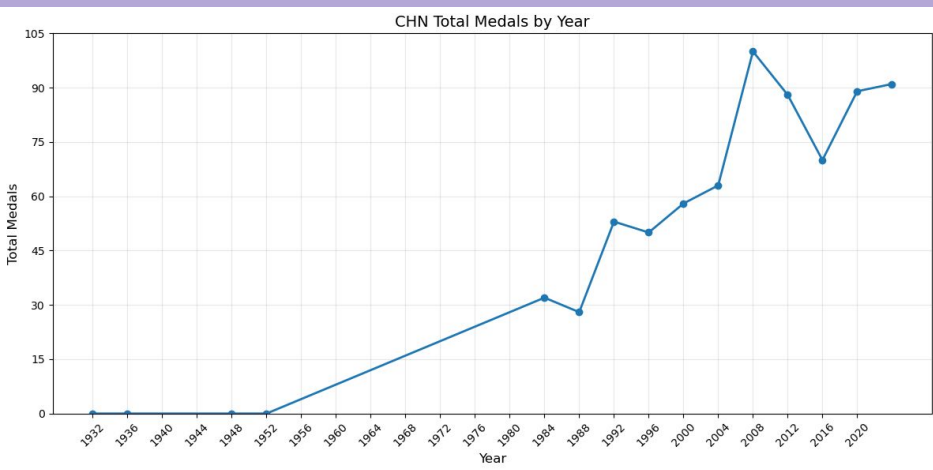
coeff. **con** MACRO
(coef. **senza**
MACRO)

*p-value < **0.10**

p-value < **0.05

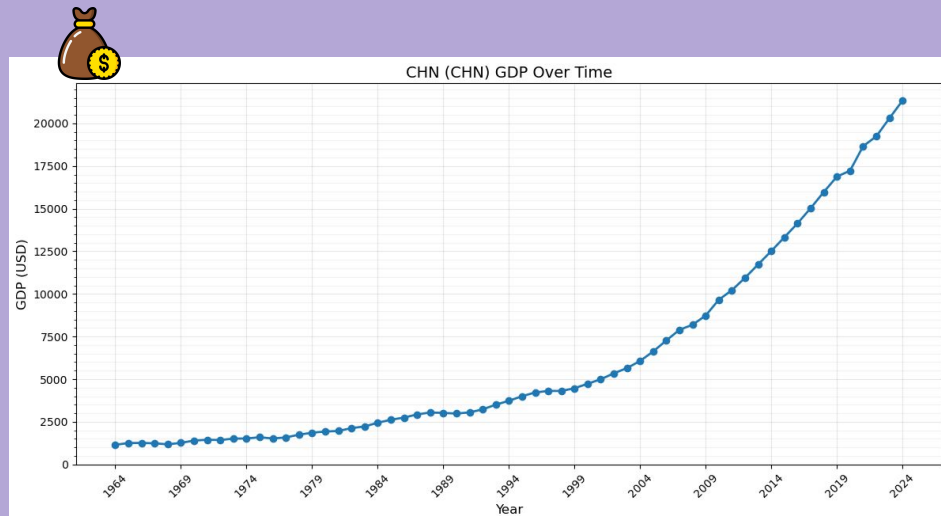
***p-value < **0.01**

OLS (AM, Top40):

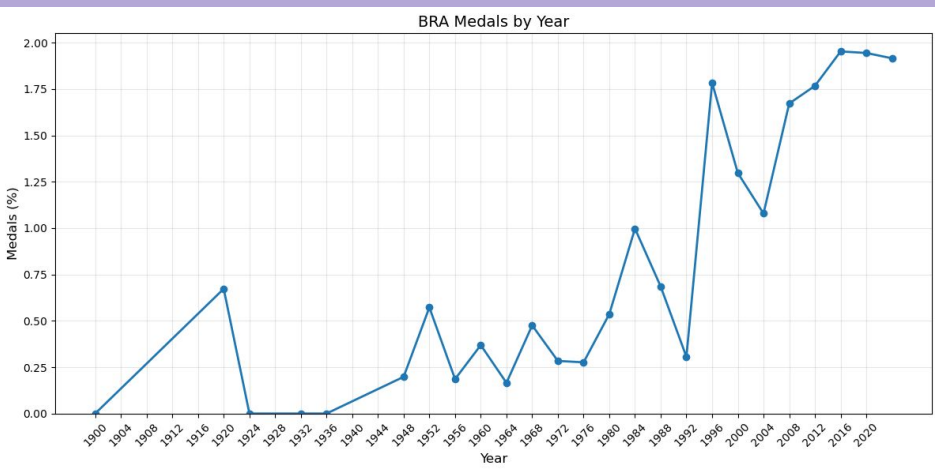


	Medaglie 
2000	6.27% (58)
2004	6.80% (63)
2008	10.45% (100)
2012	9.15% (88)

	OLS +MACRO	OLS
OG2008	5.128***	7.828***
PRE2004	1.739***	4.4421***
POST2008	3.449***	6.122***

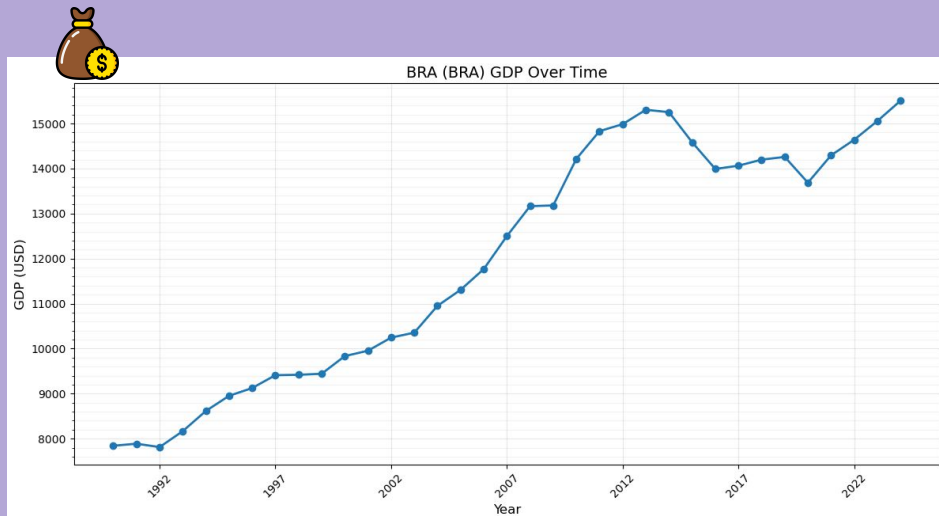


OLS (AM, Top40):



	Medaglie 
2008	1.67% (16)
2012	1.95% (17)
2016	1.77% (19)
2020	1.94% (21)

	OLS AM+MACRO	OLS +AM	OLS +MACRO
OG2016	-0.500**	0.889***	-2.027***
PRE2012	-0.632***	0.738***	-2.172***
POST2020	-0.546***	0.845***	-1.945***

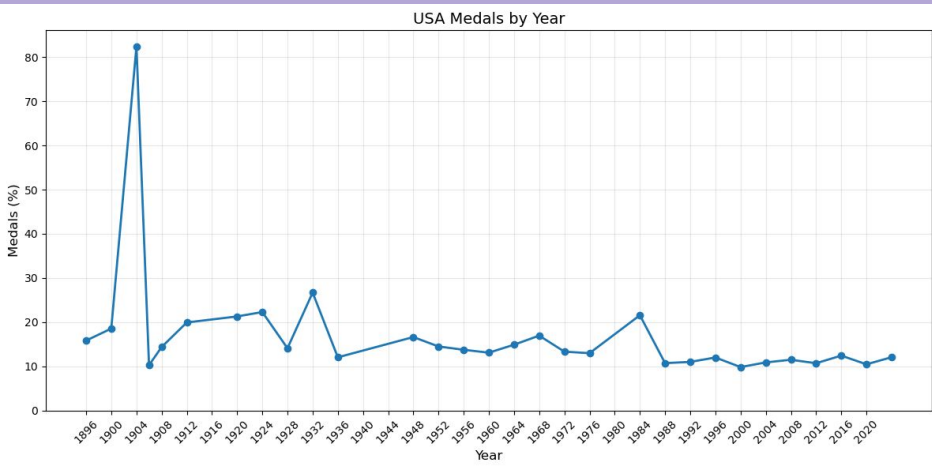


OLS

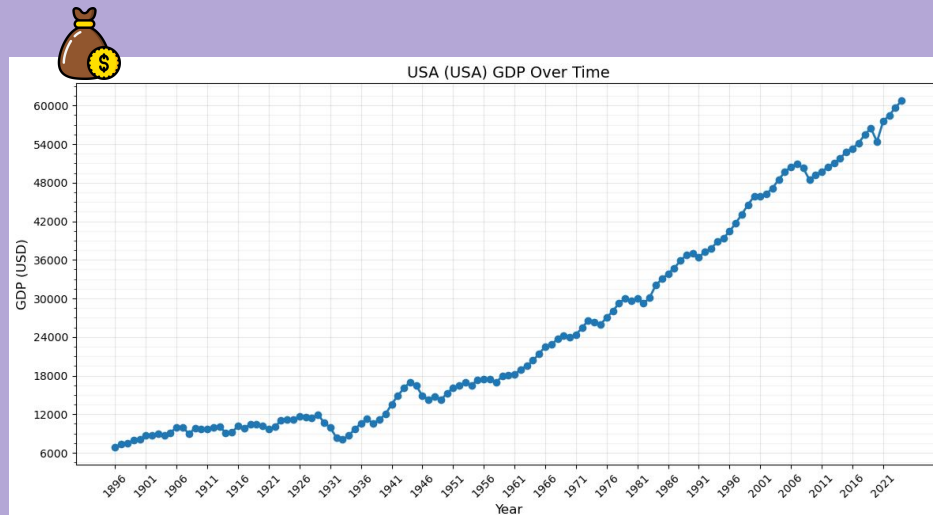
(AM, Top40):



	OLS AM+MACRO	OLS +AM	OLS +MACRO	OLS
OG1996	-0.656*	-0.255	6.403***	10.031 ***



	Medaglie 
1992	11.01% (108)
1996	12.01% (101)

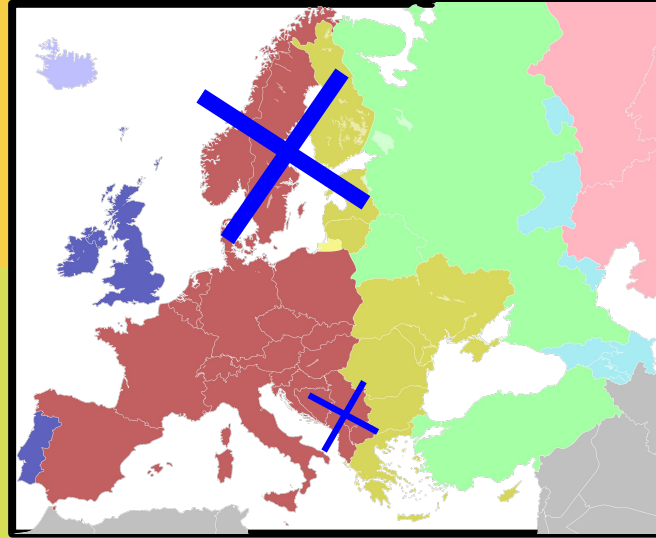


VARIABILI (CLOSE)

GMT1

Main european regions

- following actual state borders
- following cultural proximities



WEST,
CENTRAL

WIDE,
MAIN

	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996	
	0.554***	0.506***	0.497***	0.498***	0.495***	0.463***	AM (History) 
	-	0.138	0.280***	0.106	0.124	0.268***	ln_GDP 
	-	0.503***	0.547***	0.531***	0.537***	0.631***	ln_Population 
	-	0.831***	1.145***	1.009***	1.031***	0.696***	Is_Communist 
OG1900	29.395***	28.719***	-	-	-	-	
OG1904	73.601***	72.933***	-	-	-	-	
OG1908	39.000***	37.945***	-	-	-	-	
OG1912	17.399***	17.685***	-	-	-	-	
OG1920	5.675***	5.756***	-	-	-	-	
OG1924	1.100	0.825	-	-	-	-	
OG1928	3.943***	3.961***	-	-	-	-	
OG1932	12.783***	12.676***	12.613***	-	-	-	
OG1936	17.060***	16.484***	16.427***	-	-	-	
OG1948	-2.189***	-2.552***	-2.612***	-	-	-	
OG1952	0.795**	1.723***	1.743***	-	-	-	
OG1956	5.629***	5.820***	5.701***	-	-	-	
OG1960	4.465***	3.922***	3.844***	-	-	-	
OG1964	3.420***	2.546***	2.479***	2.447***	2.475***	-	
OG1968	1.012***	0.499	0.492*	0.394	0.390	-	
OG1976	0.530*	0.410	0.366	0.400	0.383	-	
OG1984	9.734***	9.395***	9.340***	9.421***	9.446***	-	
OG1988	3.359***	3.121***	3.177***	3.061***	3.113***	-	
OG1992	1.611***	1.351***	1.333***	1.317***	1.303***	-	
OG1996	0.513	0.179	0.158	0.321	0.317	0.310	
OG2000	4.183***	4.438***	4.413***	4.494***	4.503***	4.183***	
OG2004	-0.384	0.202	0.286	0.303	0.313	0.053	
OG2008	8.053***	5.591***	5.337***	5.271***	5.274***	5.171***	
OG2012	2.544***	2.398***	2.408***	2.436***	2.409***	2.141***	
OG2016	0.986***	0.120	0.165	0.027	0.029	-0.414*	
OG2020	3.142***	2.505***	2.478***	2.503***	2.488***	2.087***	
OG2024	1.520**	0.888*	2.060***	2.072***	2.116***	1.857***	

OLS

+ AM

+ YEAR

+ CLOSE

(Top40)

	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
R-squared	0.874	0.890	0.813	0.787	0.787	0.802
Adj. R-squared	0.850	0.868	0.782	0.754	0.754	0.773
Log-Likelihood	-1668.7	-1611.6	-1163.5	-843.02	-841.42	-466.99
OG1900	29.395***	28.719***	-	-	-	-
OG1904	73.601***	72.933***	-	-	-	-
OG1908	39.000***	37.945***	-	-	-	-
OG1912	17.399***	17.685***	-	-	-	-
OG1920	5.675***	5.756***	-	-	-	-
OG1924	1.100	0.825	-	-	-	-
OG1928	3.943***	3.961***	-	-	-	-
OG1932	12.783***	12.676***	12.613***	-	-	-
OG1936	17.060***	16.484***	16.427***	-	-	-
OG1948	-2.189***	-2.552***	-2.612***	-	-	-
OG1952	0.795**	1.723***	1.743***	-	-	-
OG1956	5.629***	5.820***	5.701***	-	-	-
OG1960	4.465***	3.922***	3.844***	-	-	-
OG1964	3.420***	2.546***	2.479***	2.447***	2.475***	-
OG1968	1.012***	0.499	0.492*	0.394	0.390	-
OG1976	0.530*	0.410	0.366	0.400	0.383	-
OG1984	9.734***	9.395***	9.340***	9.421***	9.446***	-
OG1988	3.359***	3.121***	3.177***	3.061***	3.113***	-
OG1992	1.611***	1.351***	1.333***	1.317***	1.303***	-
OG1996	0.513	0.179	0.158	0.321	0.317	0.310
OG2000	4.183***	4.438***	4.413***	4.494***	4.503***	4.183***
OG2004	-0.384	0.202	0.286	0.303	0.313	0.053
OG2008	8.053***	5.591***	5.337***	5.271***	5.274***	5.171***
OG2012	2.544***	2.398***	2.408***	2.436***	2.409***	2.141***
OG2016	0.986***	0.120	0.165	0.027	0.029	-0.414*
OG2020	3.142***	2.505***	2.478***	2.503***	2.488***	2.087***
OG2024	1.520**	0.888*	2.060***	2.072***	2.116***	1.857***

OLS

+ AM

+ YEAR

+ CLOSE

(Top40)

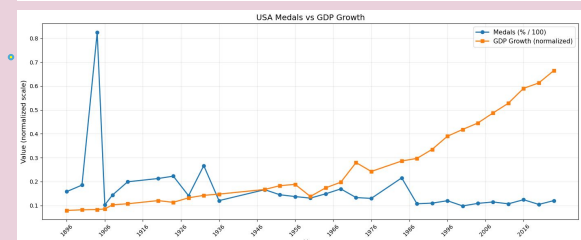
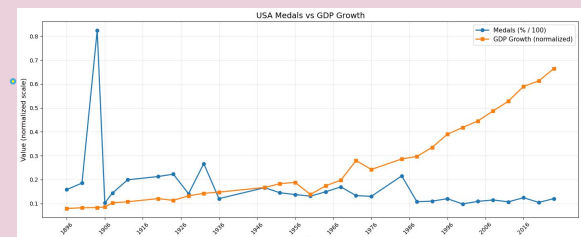
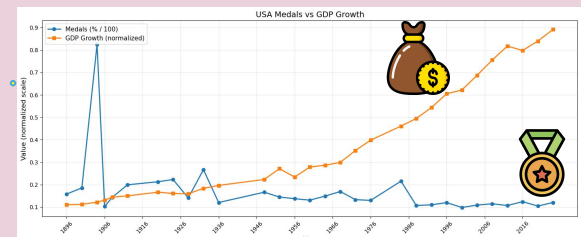
OLS (AM + YEAR + CLOSE, Top40) :

	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
OG1900	29.395***	28.719***	-	-	-	-
OG1904	73.601***	72.933***	-	-	-	-
OG1908	39.000***	37.945***	-	-	-	-
OG1912	17.399***	17.685***	-	-	-	-
OG1920	5.675***	5.756***	-	-	-	-
OG1924	1.100	0.825	-	-	-	-
OG1928	3.943***	3.961***	-	-	-	-
OG1932	12.783***	12.676***	12.613***	-	-	-
OG1936	17.060***	16.484***	16.427***	-	-	-
OG1948	-2.189***	-2.552***	-2.612***	-	-	-
OG1952	0.795**	1.723***	1.743***	-	-	-
OG1956	5.629***	5.820***	5.701***	-	-	-
OG1960	4.465***	3.922***	3.844***	-	-	-
OG1964	3.420***	2.546***	2.479***	2.447***	2.475***	-
OG1968	1.012***	0.499	0.492*	0.394	0.390	-
OG1976	0.530*	0.410	0.366	0.400	0.383	-
OG1984	9.734***	9.395***	9.340***	9.421***	9.446***	-
OG1988	3.359***	3.121***	3.177***	3.061***	3.113***	-
OG1992	1.611***	1.351***	1.333***	1.317***	1.303***	-
OG1996	0.513	0.179	0.158	0.321	0.317	0.310
OG2000	4.183***	4.438***	4.413***	4.494***	4.503***	4.183***
OG2004	-0.384	0.202	0.286	0.303	0.313	0.053
OG2008	8.053***	5.591***	5.337***	5.271***	5.274***	5.171***
OG2012	2.544***	2.398***	2.408***	2.436***	2.409***	2.141***
OG2016	0.986***	0.120	0.165	0.027	0.029	-0.414*
OG2020	3.142***	2.505***	2.478***	2.503***	2.488***	2.087***
OG2024	1.520**	0.888*	2.060***	2.072***	2.116***	1.857***

LAG 0

LAG 7

LAG 28



1900 w/o Macro

1900

1932

1964

1964 (Lag 7)

1996

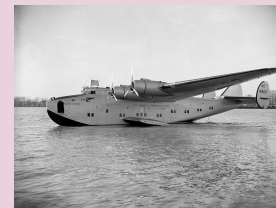
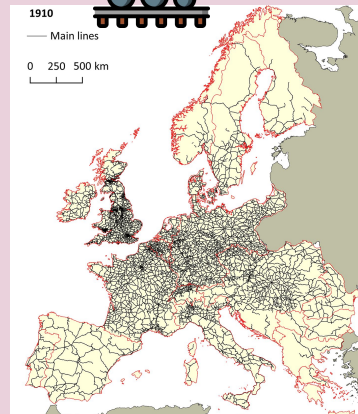
82.4%



OG1900	29.395***	28.719***	-	-	-	-
OG1904	73.601***	72.933***	-	-	-	-
OG1908	39.000***	37.945***	-	-	-	-
OG1912	17.399***	17.685***	-	-	-	-
OG1920	5.675***	5.756***	-	-	-	-
OG1924	1.100	0.825	-	-	-	-
OG1928	3.943***	3.961***	-	-	-	-
OG1932	12.783***	12.676***	12.613***	-	-	-
OG1936	17.060***	16.484***	16.427***	-	-	-
OG1948	-2.189***	-2.552***	-2.612***	-	-	-
OG1952	0.795**	1.723***	1.743***	-	-	-
OG1956	5.629***	5.820***	5.701***	-	-	-
OG1960	4.465***	3.922***	3.844***	-	-	-
OG1964	3.420***	2.546***	2.479***	2.447***	2.475***	-
OG1968	1.012***	0.499	0.492*	0.394	0.390	-
OG1976	0.530*	0.410	0.366	0.400	0.383	-
OG1984	9.734***	9.395***	9.340***	9.421***	9.446***	-
OG1988	3.359***	3.121***	3.177***	3.061***	3.113***	-
OG1992	1.611***	1.351***	1.333***	1.317***	1.303***	-
OG1996	0.513	0.179	0.158	0.321	0.317	0.310
OG2000	4.183***	4.438***	4.413***	4.494***	4.503***	4.183***
OG2004	-0.384	0.202	0.286	0.303	0.313	0.053
OG2008	8.053***	5.591***	5.337***	5.271***	5.274***	5.171***
OG2012	2.544***	2.398***	2.408***	2.436***	2.409***	2.141***
OG2016	0.986***	0.120	0.165	0.027	0.029	-0.414*
OG2020	3.142***	2.505***	2.478***	2.503***	2.488***	2.087***
OG2024	1.520**	0.888*	2.060***	2.072***	2.116***	1.857***

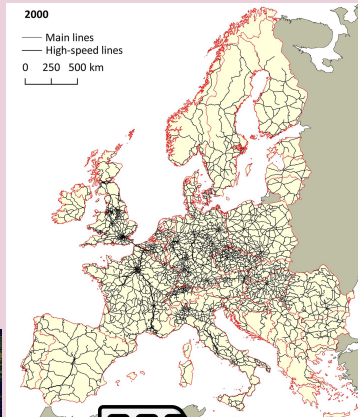


1910

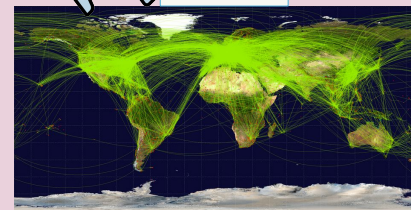
 — Main lines
 0 250 500 km


1939

2000

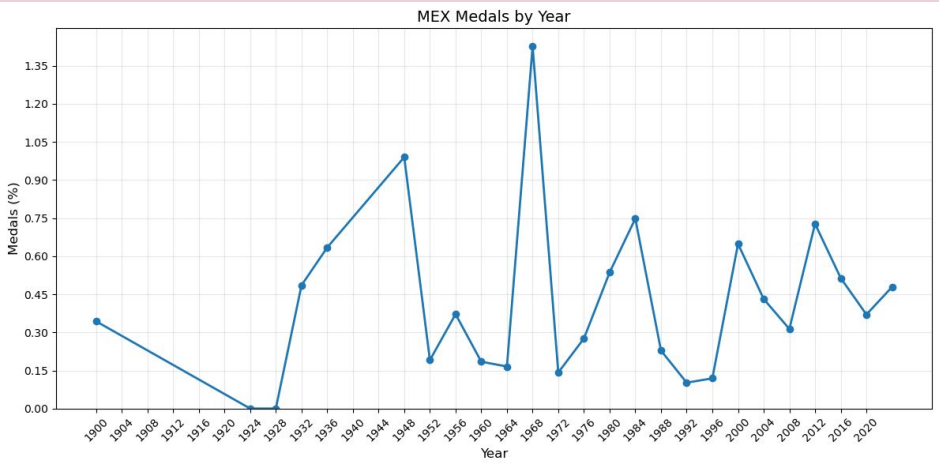
 — Main lines
 — High-speed lines
 0 250 500 km


2009



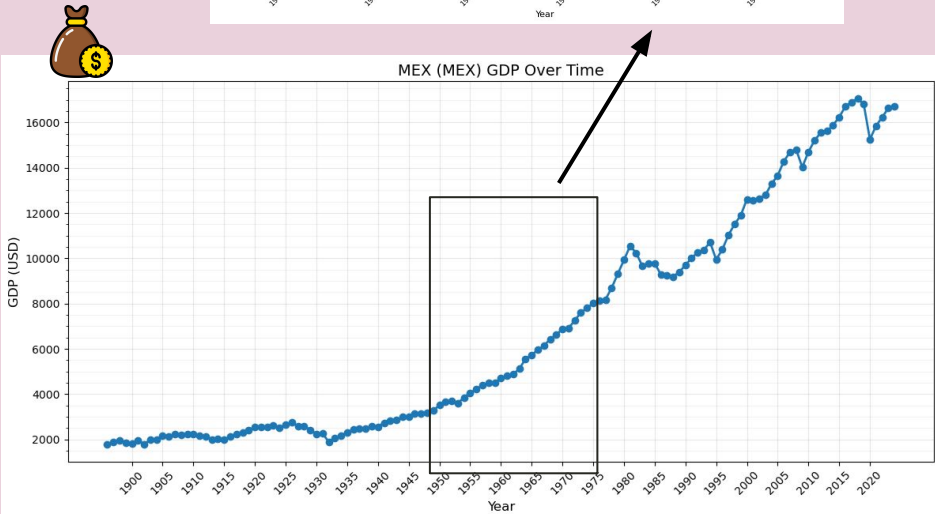
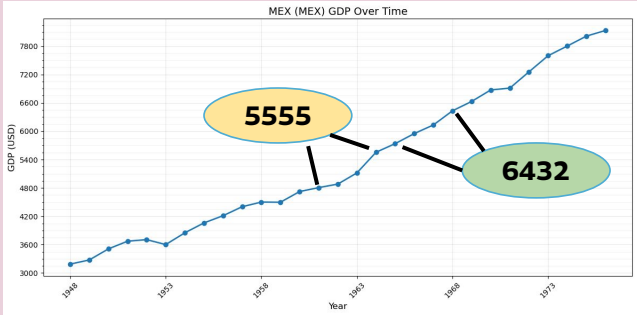
	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
OG1900	29.395***	28.719***	-	-	-	-
OG1904	73.601***	72.933***	-	-	-	-
OG1908	39.000***	37.945***	-	-	-	-
OG1912	17.399***	17.685***	-	-	-	-
OG1920	5.675***	5.756***	-	-	-	-
 OG1924	1.100	0.825	-	-	-	-
OG1928	3.943***	3.961***	-	-	-	-
OG1932	12.783***	12.676***	12.613***	-	-	-
OG1936	17.060***	16.484***	16.427***	-	-	-
OG1948	-2.189***	-2.552***	-2.612***	-	-	-
OG1952	0.795**	1.723***	1.743***	-	-	-
OG1956	5.629***	5.820***	5.701***	-	-	-
OG1960	4.465***	3.922***	3.844***	-	-	-
 OG1964	3.420***	2.546***	2.479***	2.447***	2.475***	-
OG1968	1.012***	0.499	0.492*	0.394	0.390	-
 OG1976	0.530*	0.410	0.366	0.400	0.383	-
OG1984	9.734***	9.395***	9.340***	9.421***	9.446***	-
OG1988	3.359***	3.121***	3.177***	3.061***	3.113***	-
OG1992	1.611***	1.351***	1.333***	1.317***	1.303***	-
 OG1996	0.513	0.179	0.158	0.321	0.317	0.310
OG2000	4.183***	4.438***	4.413***	4.494***	4.503***	4.183***
 OG2004	-0.384	0.202	0.286	0.303	0.313	0.053
OG2008	8.053***	5.591***	5.337***	5.271***	5.274***	5.171***
OG2012	2.544***	2.398***	2.408***	2.436***	2.409***	2.141***
 OG2016	0.986***	0.120	0.165	0.027	0.029	-0.414*
OG2020	3.142***	2.505***	2.478***	2.503***	2.488***	2.087***
OG2024	1.520**	0.888*	2.060***	2.072***	2.116***	1.857***

OLS



	Medaglie 
1964	0.17% (1)
1968	1.43% (9)

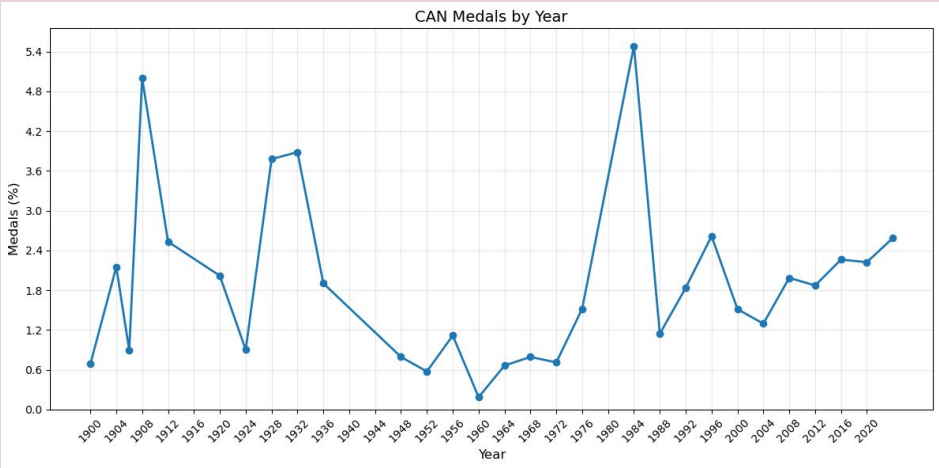
	OLS 1900 + MACRO	OLS 1900
OG1968	0.499	1.012***



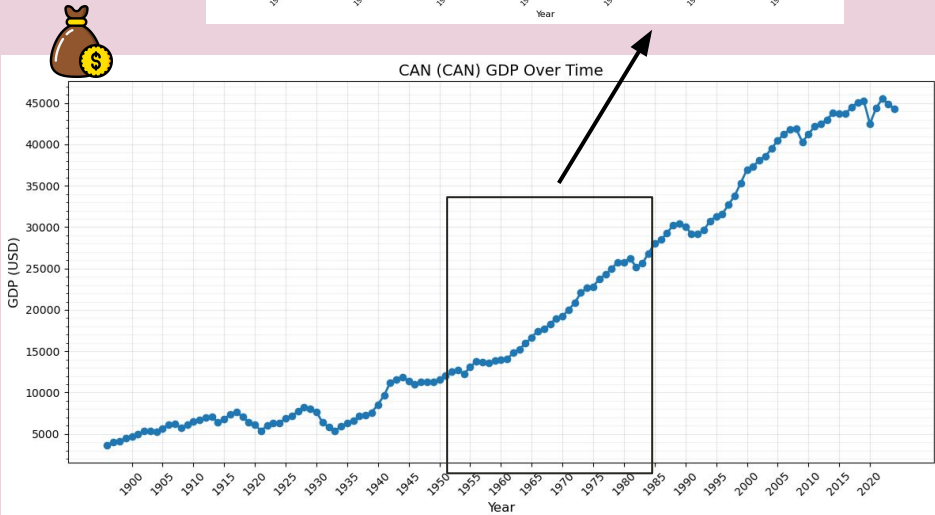
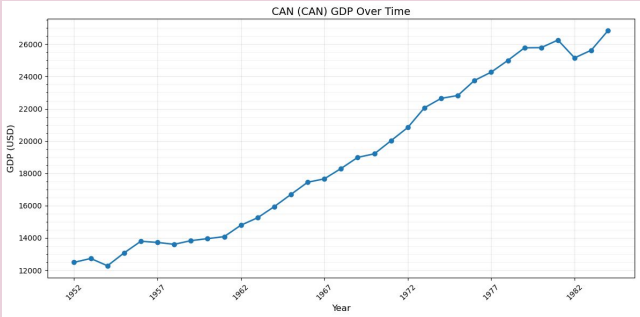
OLS



	OLS 1900 + MACRO	OLS 1900
OG1976	0.410	0.530*



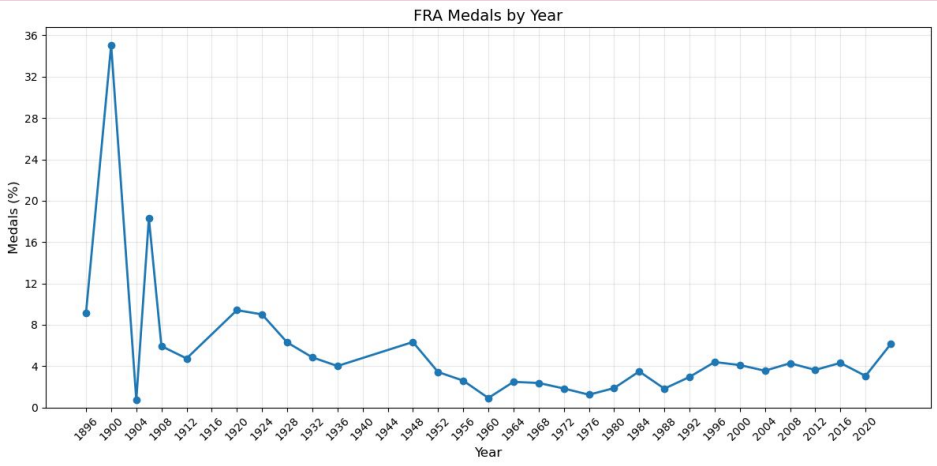
	Medaglie 
1972	0.71% (5)
1976	1.52% (11)



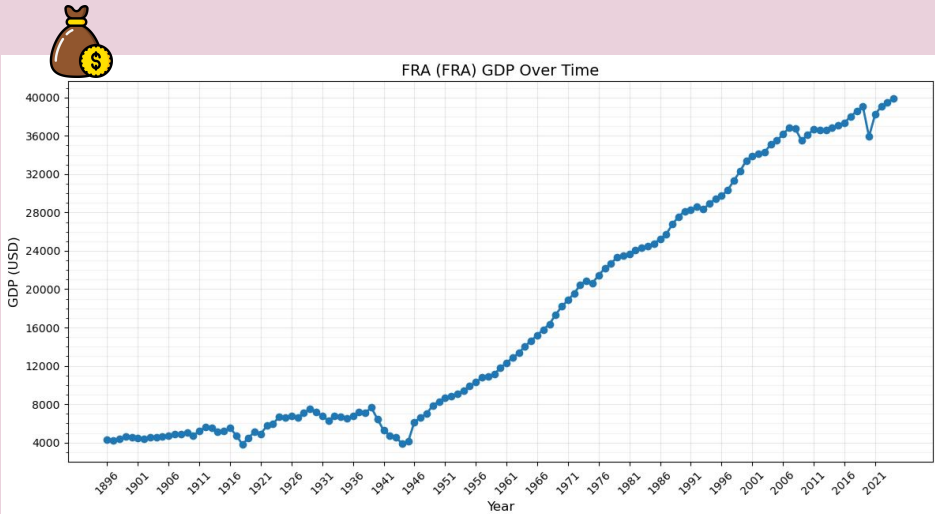
OLS



	OLS +MACRO	OLS
OG1924	0.825	1.100



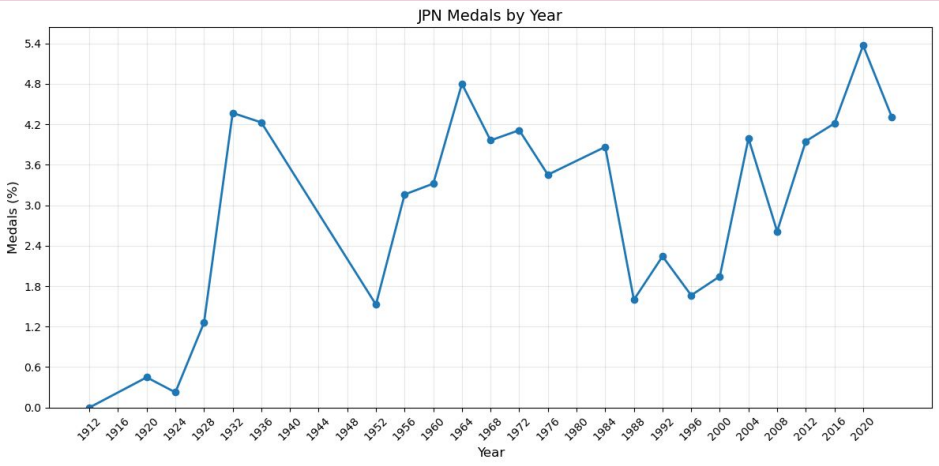
	Medaglia🏅
1920	9.42% (42)
1924	9.01% (40)



OLS

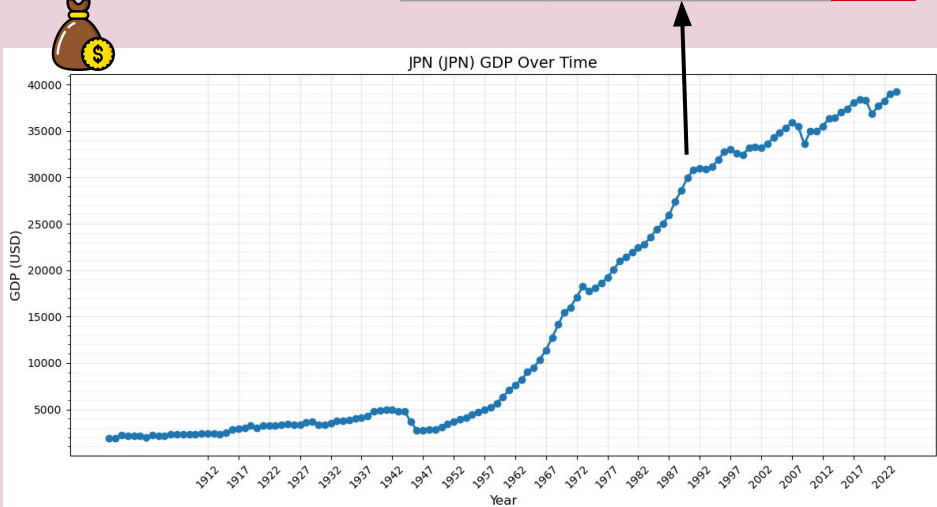


	OLS 1900	OLS 1900 +MACRO	OLS 1964 +MACRO	OLS 1996 +MACRO
OG2020	3.142***	2.505***	2.503***	2.087***



	Medaglie
1988	1.60% (14)
2016	4.21% (41)
2020	5.37% (58)

#POS (1990)	\$1B		
1	6,963	USA	
2	3,186	JPN	
3	2,660	URS	



	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
CLOSE_CENTER1936	1.320	-0.854	-0.854	-	-	-
CLOSE_CENTER1972	2.124**	2.234***	2.234***	2.847***	2.840***	-

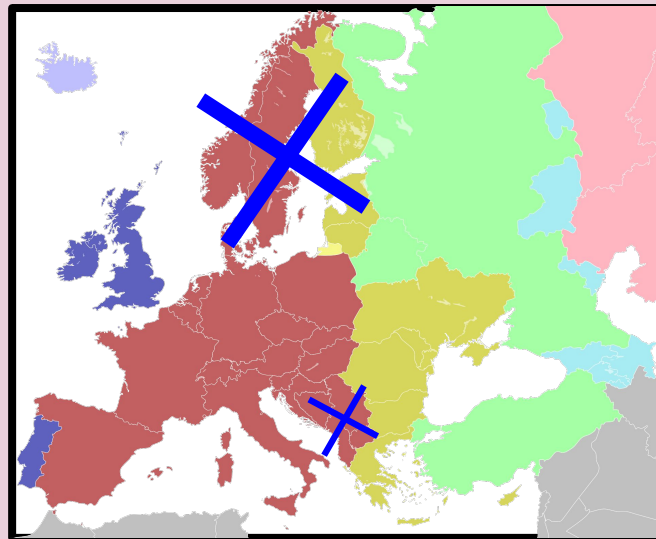


	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
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CLOSE_WEST_1900	4.944***	5.132***	-	-	-	-
CLOSE_WEST_1920	-0.618	0.023	-	-	-	-
CLOSE_WEST_1924	0.570***	0.800***	-	-	-	-
CLOSE_WEST_1928	-3.787***	-3.154***	-	-	-	-
CLOSE_WEST_2024	1.275*	1.407**	2.329***	2.497***	2.479***	2.299***



	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
CLOSE_GMT1_1900	-10.928***	-10.453***	-	-	-	-
CLOSE_GMT1_1920	4.170	3.389	-	-	-	-
CLOSE_GMT1_1924	2.261	1.876	-	-	-	-
CLOSE_GMT1_1928	5.055***	4.463***	-	-	-	-
CLOSE_GMT1_1936	3.620***	3.288**	4.908***	-	-	-
CLOSE_GMT1_1960	2.440***	2.349***	2.518***	-	-	-
CLOSE_GMT1_1972	1.174**	0.943	0.838	0.869	0.857	-
CLOSE_GMT1_1992	1.813***	1.828***	1.945***	2.073***	2.051***	-
CLOSE_GMT1_2024	-1.195	-1.197	-1.280	-1.372	-1.365*	-1.292



	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
CLOSE_WIDE_1900	9.700***	8.871***	-	-	-	-
CLOSE_WIDE_1908	-2.478**	-2.976***	-	-	-	-
CLOSE_WIDE_1920	-3.665	-3.500*	-	-	-	-
CLOSE_WIDE_1924	-3.442**	-3.278**	-	-	-	-
CLOSE_WIDE_1928	-3.821***	-3.727***	-	-	-	-
CLOSE_WIDE_1936	-3.597***	-3.012***	-2.909***	-	-	-
CLOSE_WIDE_1948	-0.565	-0.663	-0.525	-	-	-
CLOSE_WIDE_1960	-1.759***	-1.715***	-1.715***	-	-	-
CLOSE_WIDE_1972	-2.294***	-2.241***	-2.196***	-2.183***	-2.176***	-
CLOSE_WIDE_1992	-2.056***	-1.859***	-1.823***	-1.841***	-1.859***	-
CLOSE_WIDE_2004	-0.303	-0.220	-0.163	-0.114	-0.118	-0.413
CLOSE_WIDE_2012	-0.446	-0.319	-0.277	-0.229	-0.246	-0.500*
CLOSE_WIDE_2024	-0.392	-0.581	0.654	0.612	0.662	0.482



	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
CLOSE_MAIN_1900	0.098	-0.318	-	-	-	-
CLOSE_MAIN_1924	0.834	0.507	-	-	-	-
CLOSE_MAIN_1936	0.456	-0.114	-1.852	-	-	-
CLOSE_MAIN_1960	-0.734	-1.121	-1.347	-	-	-
CLOSE_MAIN_1992	0.910	0.417	0.228	0.161	0.176	-
CLOSE_MAIN_2024	0.871	0.397	0.408	0.552	0.534	0.336



Limitazioni:

- quantità di dati
- variabilità nel tempo
 - avvenimenti, guerre



Conclusioni(?)



GRAZIE per l'attenzione :)

Credits:

icone - [Flaticon.com](https://flaticon.com)

immagini - Wikipedia

paper - G. Csurilla and I. Fertő, "The less obvious effect of hosting the Olympics on sporting performance," Scientific Reports, vol. 13, no. 1, p. 819, Feb. 2023.



Appendice

	Csurilla': ZINB	ZINB (YEAR)		OLS Top 40		
Variables	All	All	Top 40	YEAR	AM+YEAR	AM
R-squared	-	-	-	0.628	0.798	0.795
Adj. R-squared	-	-	-	(0.233)	(0.706)	(0.704)
	-	-	-	0.584	0.773	0.776
	-	-	-	(0.151)	(0.674)	(0.679)
Pseudo R-squared	-	0.1580	0.1400	-	-	-
	-	(0.01514)	(0.01953)	-	-	-
Log-Likelihood	-	-2902.7	-1096.3	-566.04	-470.41	-472.70
	-	(-2857.8)	(-1249.8)	(-679.32)	(-528.94)	(-530.29)
OG1996	0.021	0.526***	0.538***	6.403***	0.354	0.656*
	(0.544***)	(3.171***)	(1.802***)	(10.031***)	(-0.575)	(-0.255)
OG2000	0.361***	1.706***	1.129***	3.850***	4.201***	4.377***
	(0.476***)	(2.496***)	(1.154***)	(4.294***)	(4.076***)	(4.211***)
OG2004	0.384	1.295***	0.321***	0.262	0.160	0.183**
	(0.273)	(1.160***)	(-0.186)	(-0.354)	(-0.372)	(-0.419***)
OG2008	0.101	-0.114	0.206	4.100***	5.186***	5.128***
	(0.459***)	(3.002***)	(1.673***)	(8.492***)	(7.897***)	(7.828***)
OG2012	0.577***	1.408***	0.915***	3.327***	2.282***	2.182***
	(0.759***)	(2.587***)	(1.264***)	(4.849***)	(2.325***)	(2.180***)
OG2016	0.362	0.255	-0.489***	-2.027***	-0.403*	-0.500***
	(0.310)	(1.289***)	(-0.039)	(-0.078)	(0.956***)	(0.889***)
OG2020	0.114	0.345***	0.406***	1.217***	2.109***	2.025***
	(0.496**)	(1.629***)	(0.951***)	(3.297***)	(3.048***)	(2.987***)
OG2024	—	0.834***	0.819***	2.686***	1.913***	1.867***
	—	(1.755***)	(1.079***)	(4.048***)	(2.031***)	(2.056***)
PRE1996	0.181	1.507***	0.895***	2.463***	2.765***	3.096***
	(0.244)	(2.269***)	(0.901***)	(2.897***)	(2.628***)	(2.893***)
PRE2000	0.446	1.063***	0.112	0.087	-0.255	-0.109
	(0.347)	(1.000***)	(-0.340**)	(-0.571*)	(-0.900***)	(-0.765***)
PRE2004	-0.454*	-0.280	-0.172	0.499	1.744***	1.739***
	(-0.295)	(2.532***)	(1.183***)	(4.717***)	(4.467***)	(4.421***)
PRE2008	0.068	0.942***	0.552***	1.505***	0.456**	0.428***
	(0.205)	(2.267***)	(0.940***)	(3.059***)	(0.466**)	(0.407***)
PRE2012	0.387	0.164	-0.542***	-2.172***	-0.511**	-0.632***
	(0.331)	(1.245***)	(-0.076)	(-0.141)	(0.901***)	(0.738***)
PRE2016	-0.100	0.501***	0.102	-0.006	0.963***	0.902***
	(0.306)	(2.059***)	(0.730***)	(2.183***)	(1.932***)	(1.872***)
PRE2020	-0.169	0.164	0.109	-0.407	-1.137***	-1.233***
	(0.002)	(1.064***)	(0.388**)	(0.983**)	(-1.019***)	(-1.071***)
POST2000	-0.189	0.298*	0.330***	4.244***	-1.573***	-1.427***
	(0.375*)	(2.947***)	(1.604***)	(7.862***)	(-2.432***)	(-2.242***)
POST2004	0.233	1.624***	0.998***	2.853***	3.318***	3.367***
	(0.301)	(2.300***)	(0.952***)	(3.314***)	(3.275***)	(3.228***)
POST2008	-0.886*	-0.246*	-1.085***	-0.964***	-1.117***	-1.153***
	(-0.953*)	(-0.219)	(-1.544***)	(-1.539***)	(-1.640***)	(-1.712***)
POST2012	-0.139	-0.357*	0.009	2.621***	3.565***	3.449***
	(0.323)	(2.890***)	(1.566***)	(7.240***)	(6.275***)	(6.122***)
POST2016	0.778***	1.404***	0.880***	3.404***	2.351***	2.276***
	(0.975***)	(2.550***)	(1.220***)	(4.855***)	(2.348***)	(2.301***)
POST2020	0.464**	-0.100	-0.416***	-1.945***	-0.422*	-0.546***
	(0.375*)	(0.612**)	(-0.064)	(-0.129)	(0.914***)	(0.845***)
POST2024	—	0.129	0.218*	0.217	0.952***	0.916***
	—	(1.403***)	(0.727***)	(2.228***)	(1.842***)	(1.858***)
ln_GDP	0.416***	0.898***	0.420***	0.848***	—	0.202***
ln_Population	0.178***	0.612***	0.426***	1.102***	—	0.620***
Is_Communist	0.696***	1.335***	0.438***	0.805***	—	0.671***
AM (History)	0.596***	—	—	—	0.463***	0.466***
	(0.716***)	—	—	—	(0.613***)	(0.6132***)
Constant	-9.016***	-12.979***	-5.849***	-17.792***	-7.934***	(0.466***)
	(-1.911***)	(2.406***)	(3.079***)	(2.083***)	(0.700***)	(0.6132***)
Inflate Const	18.51***	-14.745***	-21.808***	—	—	—
	(27.85***)	(-12.564***)	(-19.298***)	—	—	—
Alpha	—	1.179***	0.218***	—	—	—
	—	(3.781***)	(0.652***)	—	—	—

REGRESSIONE:

METRICHE

coeff. **con** MACRO
(coeff. **senza** MACRO)

OG

PRE

POST

MACRO

const

- *p-value < **0.10**
- **p-value < **0.05**
- ***p-value < **0.01**

Appendice - Table I

TABLE I
ANALISI DI REGRESSIONE: CONFRONTO DEGLI EFFETTI HOST, PRE E POST CON E (SENZA) CONTROLLI MACROECONOMICI

Variables	Csurilla ¹ : ZINB	ZINB (YEAR)		OLS Top 40		
	All	All	Top 40	YEAR	AM+YEAR	AM
R-squared	-	-	-	0.628	0.798	0.795
	-	-	-	(0.233)	(0.706)	(0.704)
Adj. R-squared	-	-	-	0.584	0.773	0.776
	-	-	-	(0.151)	(0.674)	(0.679)
Pseudo R-squared	-	0.1580	0.1400	-	-	-
	-	(0.01514)	(0.01953)	-	-	-
Log-Likelihood	-	-2902.7	-1096.3	-566.04	-470.41	-472.70
	-	(-2857.8)	(-1249.8)	(-679.32)	(-528.94)	(-530.29)
OG1996	0.021	0.526***	0.538***	6.403***	0.354	0.656*
	(0.544***)	(3.171***)	(1.802***)	(10.031***)	(-0.575)	(-0.255)
OG2000	0.361**	1.706***	1.129***	3.850***	4.201***	4.377***
	(0.476***)	(2.496***)	(1.154***)	(4.294***)	(4.076***)	(4.211***)
OG2004	0.384	1.295***	0.321***	0.262	0.160	0.183**
	(0.273)	(1.160***)	(-0.186)	(-0.354)	(-0.372)	(-0.419***)
OG2008	0.101	-0.114	0.206	4.100***	5.186***	5.128***
	(0.459**)	(3.002***)	(1.673***)	(8.492***)	(7.897***)	(7.828***)
OG2012	0.577***	1.408***	0.915***	3.327***	2.282***	2.182***
	(0.759***)	(2.587***)	(1.264***)	(4.849***)	(2.325***)	(2.180***)
OG2016	0.362	0.255	-0.489***	-2.027***	-0.403*	-0.500**
	(0.310)	(1.289***)	(-0.039)	(-0.078)	(0.956***)	(0.889***)
OG2020	0.114	0.345***	0.406***	1.217***	2.109***	2.025***
	(0.496**)	(1.629***)	(0.951***)	(3.297***)	(3.048***)	(2.987***)
OG2024	—	0.834***	0.819***	2.686***	1.913***	1.867***
	—	(1.755***)	(1.079***)	(4.048***)	(2.031***)	(2.056***)

Appendice - Table I

PRE1996	0.181 (0.244)	1.507*** (2.269***)	0.895*** (0.901***)	2.463*** (2.897***)	2.765*** (2.628***)	3.096*** (2.893***)
PRE2000	0.446 (0.347)	1.063*** (1.000***)	0.112 (-0.340**)	0.087 (-0.571*)	-0.255 (-0.900***)	-0.109 (-0.765***)
PRE2004	-0.454* (-0.295)	-0.280 (2.532***)	-0.172 (1.183***)	0.499 (4.717***)	1.744*** (4.467***)	1.739*** (4.421***)
PRE2008	0.068 (0.205)	0.942*** (2.267***)	0.552*** (0.940***)	1.505*** (3.059***)	0.456** (0.466**)	0.428*** (0.407***)
PRE2012	0.387 (0.331)	0.164 (1.245***)	-0.542*** (-0.076)	-2.172*** (-0.141)	-0.511** (0.901***)	-0.632*** (0.738***)
PRE2016	-0.100 (0.306)	0.501*** (2.059***)	0.102 (0.730***)	-0.006 (2.183***)	0.963*** (1.932***)	0.902*** (1.872***)
PRE2020	-0.169 (0.002)	0.164 (1.064***)	0.109 (0.388**)	-0.407 (0.983**)	-1.137*** (-1.019***)	-1.233*** (-1.071***)
POST2000	-0.189 (0.375*)	0.298* (2.947***)	0.330*** (1.604***)	4.244*** (7.862***)	-1.573*** (-2.432***)	-1.427*** (-2.242***)
POST2004	0.233 (0.301)	1.624*** (2.300***)	0.998*** (0.952***)	2.853*** (3.314***)	3.318*** (3.275***)	3.367*** (3.228***)
POST2008	-0.886* (-0.953*)	-0.246* (-0.219)	-1.085*** (-1.544***)	-0.964*** (-1.539***)	-1.117*** (-1.640***)	-1.153*** (-1.712***)
POST2012	-0.139 (0.323)	-0.357* (2.890***)	0.009 (1.566***)	2.621*** (7.240***)	3.565*** (6.275***)	3.449*** (6.122***)
POST2016	0.778*** (0.975***)	1.404*** (2.550***)	0.880*** (1.220***)	3.404*** (4.855***)	2.351*** (2.348***)	2.276*** (2.301***)
POST2020	0.464** (0.375*)	-0.100 (0.612***)	-0.416*** (-0.064)	-1.945*** (-0.129)	-0.422* (0.914***)	-0.546*** (0.845***)
POST2024	— —	0.129 (1.403***)	0.218* (0.727***)	0.217 (2.228***)	0.952*** (1.842***)	0.916*** (1.858***)

Appendice - Table I

ln_GDP	0.416***	0.898***	0.420***	0.848***	0.250***	0.202***
ln_Population	0.178***	0.612***	0.426***	1.102***	0.624***	0.620***
Is_Communist	0.696***	1.335***	0.438***	0.805***	0.695***	0.671***
AM (History)	0.596*** (0.716***)	— —	— —	0.463*** —	0.463*** (0.613***)	0.466*** (0.6132***)
Constant	-9.016*** (-1.911***)	-12.979*** (2.406***)	-5.849*** (3.079***)	-17.792*** (2.083***)	-7.934*** (0.700***)	(0.466***) (0.6132***)
Inflate Const	18.51** (27.85***)	-14.745*** (-12.564***)	-21.808*** (-19.298***)	— —	— —	— —
Alpha	— —	1.179*** (3.781***)	0.218*** (0.652***)	— —	— —	— —

Coefficienti con (senza) GDP/POP/COMM. YEAR usati ma omissi. Significatività: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendice - Table II

TABLE II
CONFRONTO MODELLI OLS TOP 40: EFFETTO HOST E VARIABILI MACROECONOMICHE PER DIVERSI PERIODI E LAG

Variables	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
Model Evaluation Metrics						
R-squared	0.874	0.890	0.813	0.787	0.787	0.802
Adj. R-squared	0.850	0.868	0.782	0.754	0.754	0.773
Log-Likelihood	-1668.7	-1611.6	-1163.5	-843.02	-841.42	-466.99
Baseline & Macroeconomic Controls						
Constant	1.542***	-4.624*	-7.740***	-5.737***	-6.006***	-8.151***
AM (History)	0.554***	0.506***	0.497***	0.498***	0.495***	0.463***
ln_GDP	-	0.138	0.280***	0.106	0.124	0.268***
ln_Population	-	0.503***	0.547***	0.531***	0.537***	0.631***
Is_Communist	-	0.831***	1.145***	1.009***	1.031***	0.696***

Appendice - Table II

Host Event Variables

OG1900	29.395***	28.719***	-	-	-	-
OG1904	73.601***	72.933***	-	-	-	-
OG1908	39.000***	37.945***	-	-	-	-
OG1912	17.399***	17.685***	-	-	-	-
OG1920	5.675***	5.756***	-	-	-	-
OG1924	1.100	0.825	-	-	-	-
OG1928	3.943***	3.961***	-	-	-	-
OG1932	12.783***	12.676***	12.613***	-	-	-
OG1936	17.060***	16.484***	16.427***	-	-	-
OG1948	-2.189***	-2.552***	-2.612***	-	-	-
OG1952	0.795**	1.723***	1.743***	-	-	-
OG1956	5.629***	5.820***	5.701***	-	-	-
OG1960	4.465***	3.922***	3.844***	-	-	-
OG1964	3.420***	2.546***	2.479***	2.447***	2.475***	-
OG1968	1.012***	0.499	0.492*	0.394	0.390	-
OG1976	0.530*	0.410	0.366	0.400	0.383	-
OG1984	9.734***	9.395***	9.340***	9.421***	9.446***	-
OG1988	3.359***	3.121***	3.177***	3.061***	3.113***	-
OG1992	1.611***	1.351***	1.333***	1.317***	1.303***	-
OG1996	0.513	0.179	0.158	0.321	0.317	0.310
OG2000	4.183***	4.438***	4.413***	4.494***	4.503***	4.183***
OG2004	-0.384	0.202	0.286	0.303	0.313	0.053
OG2008	8.053***	5.591***	5.337***	5.271***	5.274***	5.171***
OG2012	2.544***	2.398***	2.408***	2.436***	2.409***	2.141***
OG2016	0.986***	0.120	0.165	0.027	0.029	-0.414*
OG2020	3.142***	2.505***	2.478***	2.503***	2.488***	2.087***
OG2024	1.520**	0.888*	2.060***	2.072***	2.116***	1.857***

Appendice - Table II

Variables	1900 w/o Macro	1900	1932	1964	1964 (Lag 7)	1996
Regional Contagion Variables (CLOSE)						
CLOSE_CENTER1936	1.320	-0.854	-0.854	-	-	-
CLOSE_CENTER1972	2.124**	2.234***	2.234***	2.847***	2.840***	-
CLOSE_GMT1_1900	-10.928***	-10.453***	-	-	-	-
CLOSE_GMT1_1920	4.170	3.389	-	-	-	-
CLOSE_GMT1_1924	2.261	1.876	-	-	-	-
CLOSE_GMT1_1928	5.055***	4.463***	-	-	-	-
CLOSE_GMT1_1936	3.620***	3.288**	4.908***	-	-	-
CLOSE_GMT1_1960	2.440***	2.349***	2.518***	-	-	-
CLOSE_GMT1_1972	1.174**	0.943	0.838	0.869	0.857	-
CLOSE_GMT1_1992	1.813***	1.828***	1.945***	2.073***	2.051***	-
CLOSE_GMT1_2024	-1.195	-1.197	-1.280	-1.372	-1.365*	-1.292
CLOSE_MAIN_1900	0.098	-0.318	-	-	-	-
CLOSE_MAIN_1924	0.834	0.507	-	-	-	-
CLOSE_MAIN_1936	0.456	-0.114	-1.852	-	-	-
CLOSE_MAIN_1960	-0.734	-1.121	-1.347	-	-	-
CLOSE_MAIN_1992	0.910	0.417	0.228	0.161	0.176	-
CLOSE_MAIN_2024	0.871	0.397	0.408	0.552	0.534	0.336
CLOSE_WEST_1900	4.944***	5.132***	-	-	-	-
CLOSE_WEST_1920	-0.618	0.023	-	-	-	-
CLOSE_WEST_1924	0.570***	0.800***	-	-	-	-
CLOSE_WEST_1928	-3.787***	-3.154***	-	-	-	-
CLOSE_WEST_2024	1.275*	1.407**	2.329***	2.497***	2.479***	2.299***

Appendice - Table II

CLOSE_WIDE_1900	9.700***	8.871***	-	-	-	-
CLOSE_WIDE_1908	-2.478**	-2.976***	-	-	-	-
CLOSE_WIDE_1920	-3.665	-3.500*	-	-	-	-
CLOSE_WIDE_1924	-3.442**	-3.278**	-	-	-	-
CLOSE_WIDE_1928	-3.821***	-3.727***	-	-	-	-
CLOSE_WIDE_1936	-3.597***	-3.012***	-2.909***	-	-	-
CLOSE_WIDE_1948	-0.565	-0.663	-0.525	-	-	-
CLOSE_WIDE_1960	-1.759***	-1.715***	-1.715***	-	-	-
CLOSE_WIDE_1972	-2.294***	-2.241***	-2.196***	-2.183***	-2.176***	-
CLOSE_WIDE_1992	-2.056***	-1.859***	-1.823***	-1.841***	-1.859***	-
CLOSE_WIDE_2004	-0.303	-0.220	-0.163	-0.114	-0.118	-0.413
CLOSE_WIDE_2012	-0.446	-0.319	-0.277	-0.229	-0.246	-0.500*
CLOSE_WIDE_2024	-0.392	-0.581	0.654	0.612	0.662	0.482

Note: PRE, POST e YEAR dummy rimosse per chiarezza. Significatività: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendice - Fig 2.1

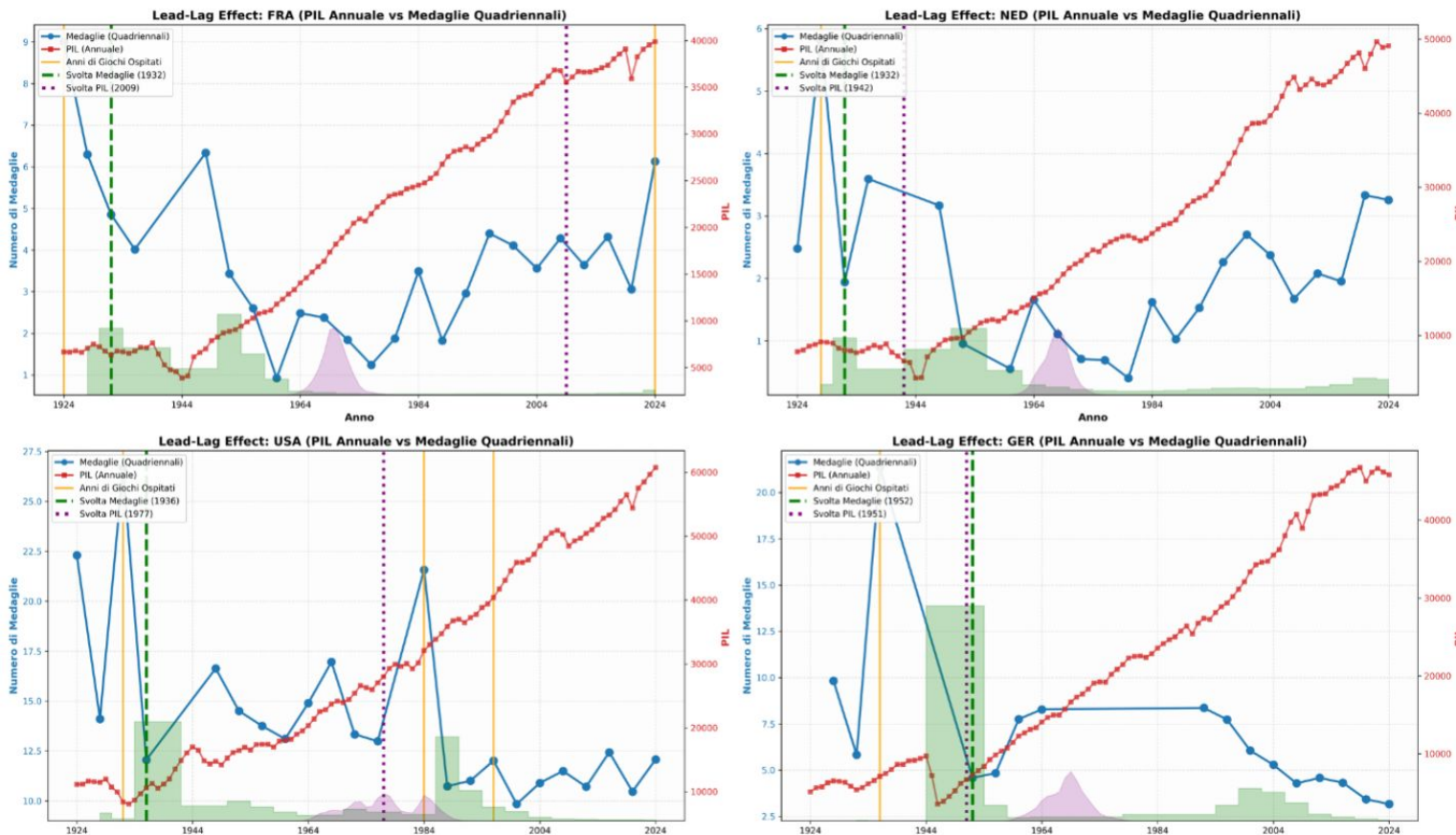


Fig. 2. Changepoint Detection per PIL e Medagliere in Nazioni Ospitanti(1).
La distribuzione di probabilità a posteriori normalizzata è visualizzata come area ombreggiata.

Appendice - Fig 2.1

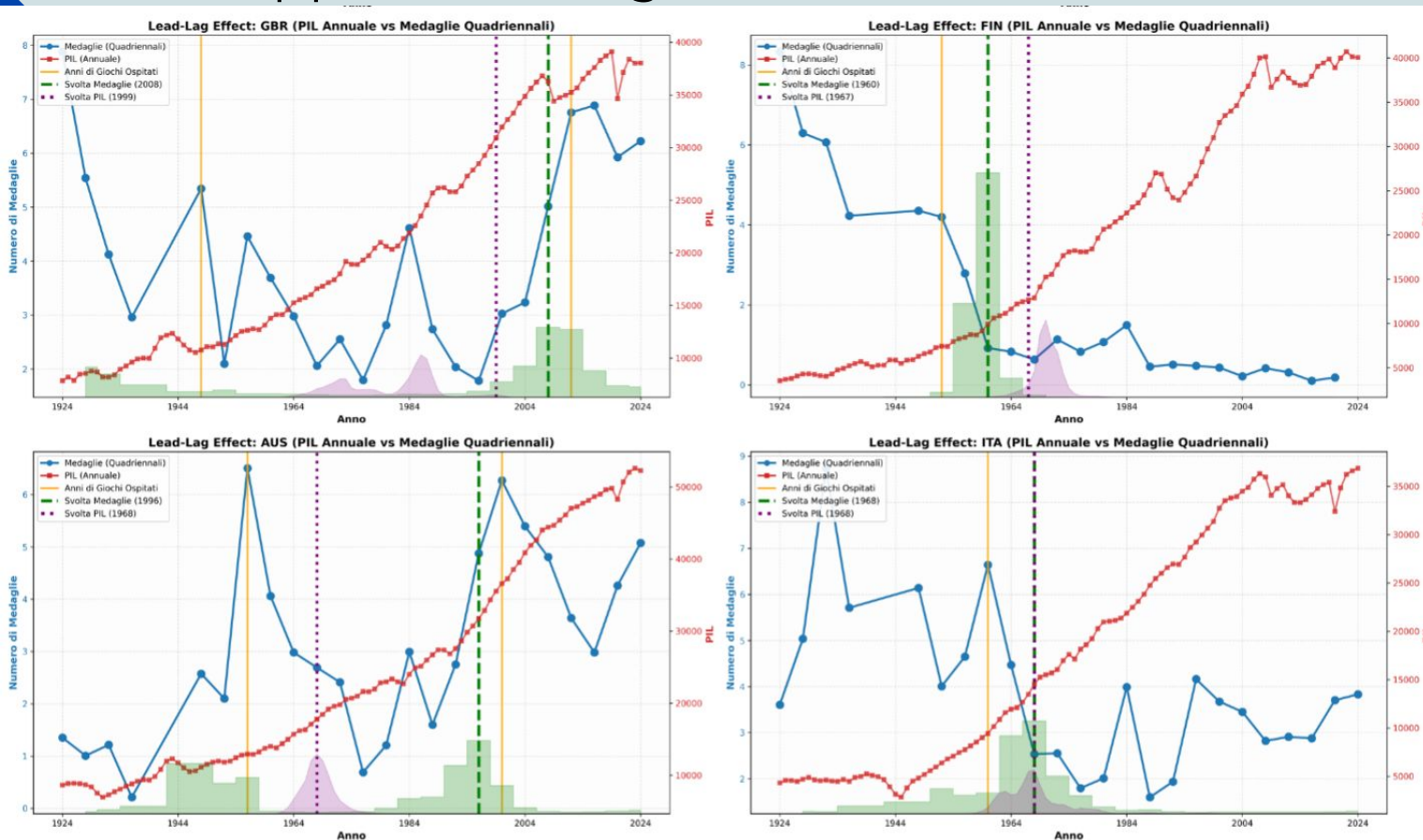


Fig. 2. Changepoint Detection per PIL e Medagliere in Nazioni Ospitanti(1).
La distribuzione di probabilità a posteriori normalizzata è visualizzata come area ombreggiata.

Appendice - Fig 2.2

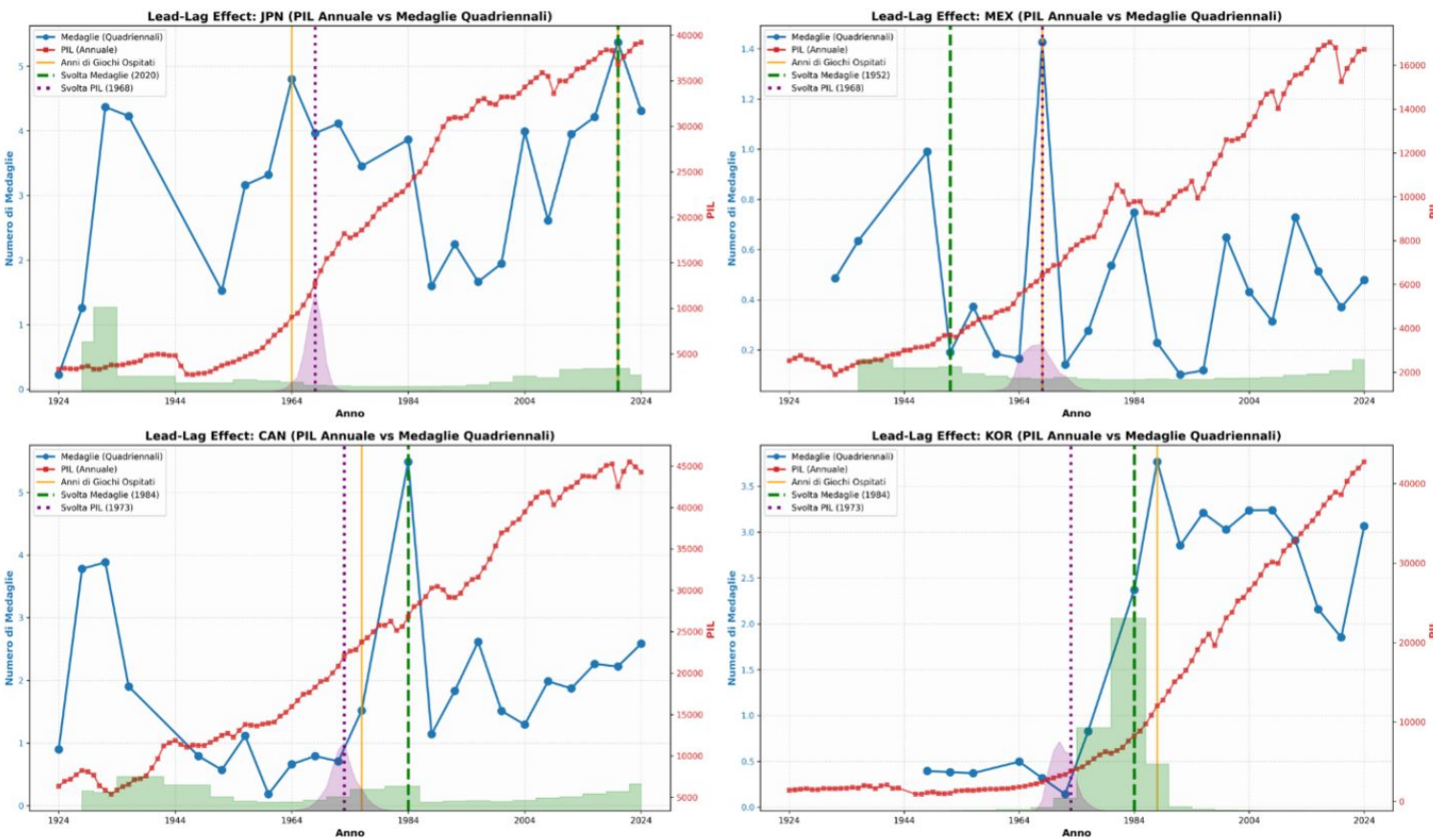


Fig. 3. Changepoint Detection per PIL e Medagliere in Nazioni Ospitanti(2).
La distribuzione di probabilità a posteriori normalizzata è visualizzata come area ombreggiata.

Appendice - Fig 2.2

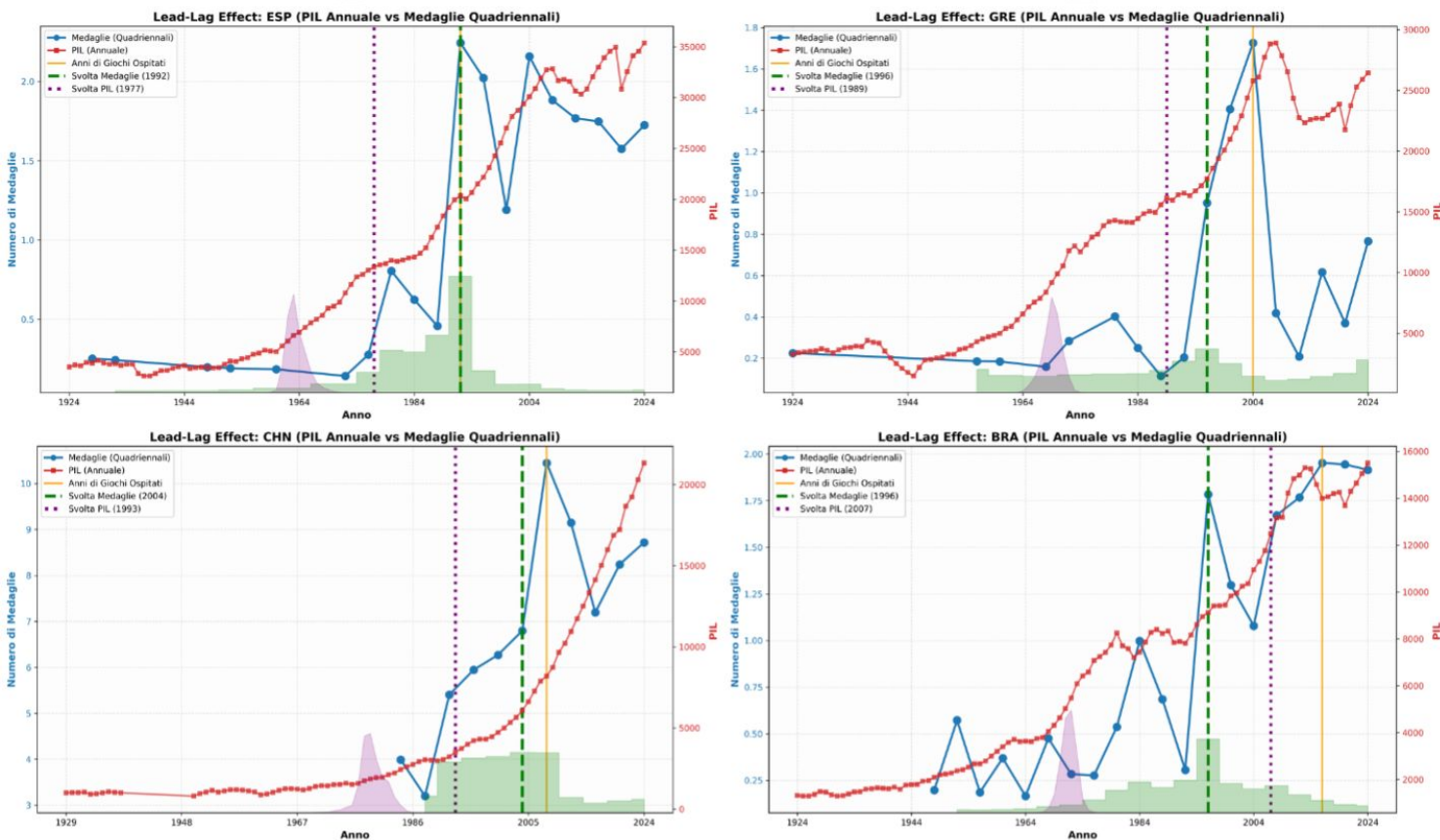


Fig. 3. Changepoint Detection per PIL e Medagliere in Nazioni Ospitanti(2).
La distribuzione di probabilità a posteriori normalizzata è visualizzata come area ombreggiata.

Appendice - host



1	Year,Season,City,Host NOC
2	1896,Summer,Athens,GRE
3	1900,Summer,Paris,FRA
4	1904,Summer,St. Louis,USA
5	1908,Summer,London,GBR
6	1912,Summer,Stockholm,SWE
7	1920,Summer,Antwerp,BEL
8	1924,Summer,Paris,FRA
9	1924,Winter,Chamonix,FRA
10	1928,Summer,Amsterdam,NED
11	1928,Winter,St. Moritz,SUI
12	1932,Summer,Los Angeles,USA
13	1932,Winter,Lake Placid,USA
14	1936,Summer,Berlin,GER
15	1936,Winter,Garmisch-Partenkirchen,GER
16	1948,Summer,London,GBR
17	1948,Winter,St. Moritz,SUI
18	1952,Summer,Helsinki,FIN
19	1952,Winter,Oslo,NOR
20	1956,Summer,Melbourne,AUS
21	1956,Winter,Cortina d'Ampezzo,ITA
22	1960,Summer,Rome,ITA
23	1960,Winter,Squaw Valley,USA
24	1964,Summer,Tokyo,JPN
25	1964,Winter,Innsbruck,AUT
26	1968,Summer,Mexico City,MEX
27	1968,Winter,Grenoble,FRA
28	1972,Summer,Munich,FRG

29	1972,Winter,Sapporo,JPN
30	1976,Summer,Montreal,CAN
31	1976,Winter,Innsbruck,AUT
32	1980,Summer,Moscow,URS
33	1980,Winter,Lake Placid,USA
34	1984,Summer,Los Angeles,USA
35	1984,Winter,Sarajevo,YUG
36	1988,Summer,Seoul,KOR
37	1988,Winter,Calgary,CAN
38	1992,Summer,Barcelona,ESP
39	1992,Winter,Albertville,FRA
40	1994,Winter,Lillehammer,NOR
41	1996,Summer,Atlanta,USA
42	1998,Winter,Nagano,JPN
43	2000,Summer,Sydney,AUS
44	2002,Winter,Salt Lake City,USA
45	2004,Summer,Athens,GRE
46	2006,Winter,Turin,ITA
47	2008,Summer,Beijing,CHN
48	2010,Winter,Vancouver,CAN
49	2012,Summer,London,GBR
50	2014,Winter,Sochi,RUS
51	2016,Summer,Rio de Janeiro,BRA
52	2018,Winter,PyeongChang,KOR
53	2020,Summer,Tokyo,JPN
54	2022,Winter,Beijing,CHN
55	2024,Summer,Paris,FRA
56	2026,Winter,Milano Cortina,ITA

Appendice - risultati OLS 1900

		coef	std err	z	P> z	[0.025	0.975]
195	const	-4.6244	2.585	-1.789	0.074	-9.692	0.443
196	Avg_Medals_History	0.5061	0.058	8.716	0.000	0.392	0.620
197	GDP	0.1382	0.173	0.799	0.424	-0.261	0.477
198	Population	0.5031	0.085	5.901	0.000	0.336	0.670
199	Is Communist	0.8314	0.194	4.294	0.000	0.452	1.211
200	YEAR1900	-0.1799	0.771	-0.233	0.816	-1.692	1.332
201	YEAR1904	-1.5080	1.185	-1.272	0.203	-3.831	0.815
202	YEAR1908	0.4796	1.103	0.435	0.664	-1.683	2.642
203	YEAR1912	0.1578	1.167	0.135	0.892	-2.130	2.446
204	YEAR1920	0.8640	1.043	0.828	0.407	-1.180	2.908
205	YEAR1924	0.2589	0.939	0.276	0.783	-1.581	2.098
206	YEAR1928	-0.1514	0.763	-0.198	0.843	-1.647	1.344
207	YEAR1932	-0.8480	0.795	-1.066	0.286	-2.407	0.711
208	YEAR1936	-0.7415	0.734	-1.011	0.312	-2.179	0.696
209	YEAR1948	0.2062	0.816	0.253	0.801	-1.393	1.806
210	YEAR1952	-1.0259	0.716	-1.432	0.152	-2.430	0.378
211	YEAR1956	-1.3415	0.671	-1.999	0.046	-2.657	-0.026
212	YEAR1960	-1.3790	0.661	-2.087	0.037	-2.674	-0.084
213	YEAR1964	-1.1763	0.663	-1.774	0.076	-2.476	0.123
214	YEAR1968	-1.2370	0.650	-1.902	0.057	-2.512	0.038
215	YEAR1972	-1.0124	0.609	-1.663	0.096	-2.206	0.181
216	YEAR1976	-1.5572	0.583	-2.669	0.008	-2.701	-0.414
217	YEAR1980	-1.2492	0.601	-2.080	0.038	-2.426	-0.072
218	YEAR1984	-0.8053	0.559	-1.441	0.150	-1.901	0.290
219	YEAR1988	-1.6531	0.530	-3.117	0.002	-2.693	-0.613
220	YEAR1992	-1.3036	0.537	-2.427	0.015	-2.356	-0.251
221	YEAR1996	-0.8874	0.530	-1.673	0.094	-1.927	0.152
222	YEAR2000	-1.0663	0.523	-2.037	0.042	-2.092	-0.040
223	YEAR2004	-1.1184	0.555	-2.017	0.044	-2.205	-0.031
224	YEAR2008	-1.2653	0.485	-2.608	0.009	-2.216	-0.314
225	YEAR2012	-1.2449	0.506	-2.458	0.014	-2.238	-0.252
226	YEAR2016	-1.3315	0.470	-2.834	0.005	-2.252	-0.411
227	YEAR2020	-1.2777	0.472	-2.705	0.007	-2.204	-0.352
228	OG1900	28.7190	0.454	63.296	0.000	27.830	29.608
229	OG1904	72.9328	1.169	62.372	0.000	70.641	75.225
230	OG1908	37.9449	0.869	43.665	0.000	36.242	39.648
231	OG1912	17.6853	0.882	20.055	0.000	15.957	19.414
232	OG1920	5.7563	0.749	7.687	0.000	4.289	7.224
233	OG1924	0.8251	0.753	1.095	0.273	-0.652	2.302
234	OG1928	3.9613	0.372	10.648	0.000	3.232	4.691
235	OG1932	12.6763	1.106	11.460	0.000	10.508	14.844
236	OG1936	16.4838	0.370	44.590	0.000	15.759	17.208
237	OG1948	-2.5521	0.567	-4.502	0.000	-3.663	-1.441
238	OG1952	1.7228	0.355	4.859	0.000	1.028	2.418
239	OG1956	5.8201	0.259	22.452	0.000	5.312	6.328
240	OG1960	3.9220	0.236	16.610	0.000	3.459	4.385
241	OG1964	2.5457	0.344	7.396	0.000	1.871	3.220
242	OG1968	0.4990	0.331	1.510	0.131	-0.149	1.147
243	OG1976	0.4103	0.303	1.356	0.175	-0.183	1.003
244	OG1984	9.3948	0.853	11.020	0.000	7.724	11.066
245	OG1988	3.1210	0.228	13.669	0.000	2.674	3.569
246	OG1992	1.3513	0.286	4.731	0.000	0.791	1.911
247	OG1996	0.1789	0.775	0.231	0.817	-1.341	1.699
248	OG2000	4.4378	0.245	18.082	0.000	3.957	4.919
249	OG2004	0.2021	0.285	0.709	0.478	-0.357	0.761
250	OG2008	5.5905	0.341	16.374	0.000	4.921	6.260
251	OG2012	2.3976	0.279	8.596	0.000	1.851	2.944
252	OG2016	0.1195	0.240	0.498	0.619	-0.351	0.590
253	OG2020	2.5046	0.248	10.115	0.000	2.019	2.990
254	OG2024	0.8879	0.469	1.893	0.058	-0.031	1.807

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255	PRE1900	8.4686	0.688	12.310	0.000	7.120	9.817
256	PRE1904	-5.2901	1.013	-5.224	0.000	-7.275	-3.305
257	PRE1908	5.3284	0.831	6.415	0.000	3.701	6.956
258	PRE1912	-1.3310	0.877	-1.517	0.129	-3.051	0.389
259	PRE1920	0.5499	0.584	0.942	0.346	-0.594	1.694
260	PRE1924	0.9652	0.104	9.273	0.000	0.761	1.169
261	PRE1928	-1.2556	1.161	-1.081	0.280	-3.532	1.021
262	PRE1932	1.0117	0.445	2.275	0.023	0.140	1.883
263	PRE1936	-1.3362	0.221	-6.039	0.000	-1.770	-0.903
264	PRE1948	0.6325	0.513	1.233	0.218	-0.373	1.638
265	PRE1952	1.1848	0.338	3.506	0.000	0.522	1.847
266	PRE1956	1.9528	0.212	9.206	0.000	1.537	2.369
267	PRE1960	1.4162	0.273	5.188	0.000	0.881	1.951
268	PRE1964	-0.7455	0.318	-2.342	0.019	-1.370	-0.122
269	PRE1972	-0.9294	0.303	-3.065	0.002	-1.524	-0.335
270	PRE1984	1.0593	0.285	3.714	0.000	0.500	1.618
271	PRE1988	-0.0428	0.259	-0.165	0.869	-0.551	0.465
272	PRE1992	-0.5638	0.800	-0.705	0.481	-2.131	1.004
273	PRE1996	2.9689	0.245	12.111	0.000	2.488	3.449
274	PRE2000	-0.1577	0.225	-0.700	0.484	-0.599	0.283
275	PRE2004	2.0486	0.443	4.627	0.000	1.181	2.916
276	PRE2008	0.6607	0.231	2.865	0.004	0.209	1.113
277	PRE2012	-0.1026	0.310	-0.331	0.741	-0.710	0.505
278	PRE2016	1.4372	0.230	6.260	0.000	0.987	1.887
279	PRE2020	-0.9414	0.222	-4.234	0.000	-1.377	-0.506
280	POST1900	-27.9510	2.014	-13.876	0.000	-31.899	-24.003
281	POST1904	-10.8416	1.460	-7.424	0.000	-13.704	-7.979
282	POST1908	-4.5443	1.673	-2.716	0.007	-7.824	-1.265
283	POST1912	2.8679	1.000	2.869	0.004	0.908	4.827
284	POST1920	9.0077	0.798	11.293	0.000	7.444	10.571
285	POST1924	-0.1650	0.085	-1.937	0.053	-0.332	0.002
286	POST1928	1.0879	0.352	3.088	0.002	0.397	1.778
287	POST1932	0.4842	0.414	1.169	0.242	-0.327	1.296
288	POST1936	-2.1975	1.128	-1.948	0.051	-4.409	0.014
289	POST1952	-4.3395	0.417	-10.405	0.000	-5.157	-3.522
290	POST1956	0.6474	0.263	2.462	0.014	0.132	1.163
291	POST1960	3.1449	0.257	12.231	0.000	2.641	3.649
292	POST1964	1.4123	0.303	4.663	0.000	0.819	2.006
293	POST1968	1.5637	0.362	4.322	0.000	0.855	2.273
294	POST1972	-1.1411	0.292	-3.909	0.000	-1.713	-0.569
295	POST1988	-0.6808	0.819	-0.831	0.406	-2.287	0.925
296	POST1992	1.6327	0.245	6.670	0.000	1.153	2.112
297	POST1996	0.6392	0.293	2.184	0.029	0.066	1.213
298	POST2000	-1.7000	0.749	-2.271	0.023	-3.167	-0.233
299	POST2004	3.4854	0.316	11.043	0.000	2.867	4.104
300	POST2008	-0.9743	0.187	-5.208	0.000	-1.341	-0.608
301	POST2012	3.8842	0.393	9.873	0.000	3.113	4.655
302	POST2016	2.5844	0.213	12.105	0.000	2.166	3.003
303	POST2020	0.0183	0.255	0.072	0.943	-0.481	0.518
304	POST2024	0.1135	0.430	0.264	0.792	-0.730	0.957

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		coef	std err	z	P> z	[0.025	0.975]
192	const	1.5416	0.477	3.234	0.001	0.607	2.476
193	Avg Medals_History	0.5537	0.049	11.350	0.000	0.458	0.649
194	YEAR1900	-0.9613	0.535	-1.798	0.072	-2.009	0.086
195	YEAR1904	-2.2274	1.211	-1.839	0.066	-4.601	0.146
196	YEAR1908	-0.5307	1.021	-0.520	0.603	-2.531	1.470
197	YEAR1912	-0.4128	1.071	-0.385	0.700	-2.513	1.687
198	YEAR1920	0.3022	0.874	0.346	0.730	-1.411	2.015
199	YEAR1924	-0.2250	0.767	-0.294	0.769	-1.727	1.277
200	YEAR1928	-0.6379	0.588	-1.084	0.278	-1.791	0.515
201	YEAR1932	-1.1063	0.665	-1.665	0.096	-2.409	0.196
202	YEAR1936	-1.0094	0.563	-1.793	0.073	-2.113	0.094
203	YEAR1948	-0.1286	0.678	-0.190	0.850	-1.458	1.201
204	YEAR1952	-1.1213	0.605	-1.853	0.064	-2.307	0.064
205	YEAR1956	-1.4938	0.583	-2.562	0.010	-2.636	-0.351
206	YEAR1960	-1.5843	0.578	-2.741	0.006	-2.717	-0.451
207	YEAR1964	-1.3011	0.621	-2.095	0.036	-2.518	-0.084
208	YEAR1968	-1.3137	0.628	-2.092	0.036	-2.544	-0.083
209	YEAR1972	-1.0564	0.597	-1.768	0.077	-2.227	0.115
210	YEAR1976	-1.4829	0.604	-2.455	0.014	-2.667	-0.299
211	YEAR1980	-1.1833	0.626	-1.891	0.059	-2.410	0.043
212	YEAR1984	-0.6054	0.622	-0.973	0.330	-1.825	0.614
213	YEAR1988	-1.4592	0.566	-2.578	0.010	-2.569	-0.350
214	YEAR1992	-1.0541	0.588	-1.793	0.073	-2.207	0.098
215	YEAR1996	-0.5498	0.589	-0.933	0.351	-1.705	0.605
216	YEAR2000	-0.6936	0.593	-1.169	0.242	-1.856	0.469
217	YEAR2004	-0.7488	0.633	-1.184	0.236	-1.989	0.491
218	YEAR2008	-0.8952	0.566	-1.582	0.114	-2.005	0.214
219	YEAR2012	-0.8536	0.598	-1.426	0.154	-2.027	0.319
220	YEAR2016	-0.9098	0.563	-1.617	0.106	-2.012	0.193
221	YEAR2020	-0.8474	0.579	-1.464	0.143	-1.982	0.287
222	OG1900	29.3953	0.449	65.406	0.000	28.514	30.276
223	OG1904	73.6014	1.350	54.535	0.000	70.956	76.247
224	OG1908	38.9960	0.936	41.655	0.000	37.161	40.831
225	OG1912	17.3989	0.879	19.789	0.000	15.676	19.122
226	OG1920	5.6754	0.676	8.394	0.000	4.350	7.001
227	OG1924	1.0998	0.752	1.462	0.144	-0.374	2.574
228	OG1928	3.9426	0.297	13.288	0.000	3.361	4.524
229	OG1932	12.7825	1.145	11.167	0.000	10.539	15.026
230	OG1936	17.0596	0.351	48.535	0.000	16.371	17.749
231	OG1948	-2.1890	0.581	-3.765	0.000	-3.329	-1.049
232	OG1952	0.7952	0.308	2.586	0.010	0.192	1.398
233	OG1956	5.6289	0.233	24.116	0.000	5.171	6.086
234	OG1960	4.4647	0.251	17.814	0.000	3.974	4.956
235	OG1964	3.4202	0.305	11.226	0.000	2.823	4.017
236	OG1968	1.0120	0.331	3.057	0.002	0.363	1.661
237	OG1976	0.5299	0.287	1.848	0.065	-0.032	1.092
238	OG1984	9.7341	0.925	10.529	0.000	7.922	11.546
239	OG1988	3.3585	0.241	13.924	0.000	2.886	3.831
240	OG1992	1.6111	0.310	5.192	0.000	1.003	2.219
241	OG1996	0.5132	0.886	0.580	0.562	-1.222	2.249
242	OG2000	4.1832	0.294	14.209	0.000	3.606	4.760
243	OG2004	-0.3842	0.361	-1.063	0.288	-1.093	0.324
244	OG2008	8.0531	0.219	36.775	0.000	7.624	8.482
245	OG2012	2.5437	0.355	7.165	0.000	1.848	3.240
246	OG2016	0.9856	0.219	4.506	0.000	0.557	1.414
247	OG2020	3.1420	0.240	13.112	0.000	2.672	3.612
248	OG2024	1.5196	0.611	2.486	0.013	0.322	2.718

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249	PRE1900	9.2226	0.753	12.276	0.000	7.733	10.685
250	PRE1904	-4.6700	1.182	-3.951	0.000	-6.986	-2.354
251	PRE1908	5.5211	0.882	6.257	0.000	3.792	7.251
252	PRE1912	-1.4101	0.879	-1.605	0.109	-3.132	0.312
253	PRE1920	0.8646	0.536	1.613	0.107	-0.186	1.916
254	PRE1924	0.9093	0.127	7.152	0.000	0.660	1.158
255	PRE1928	-0.9879	1.204	-0.820	0.412	-3.348	1.373
256	PRE1932	1.5521	0.459	3.384	0.001	0.653	2.451
257	PRE1936	-0.5467	0.309	-1.770	0.077	-1.152	0.059
258	PRE1948	-0.1108	0.436	-0.254	0.799	-0.965	0.743
259	PRE1952	0.8834	0.288	3.067	0.002	0.319	1.448
260	PRE1956	2.4061	0.227	10.618	0.000	1.962	2.850
261	PRE1960	2.3105	0.240	9.621	0.000	1.840	2.781
262	PRE1964	-0.2719	0.325	-0.838	0.402	-0.908	0.364
263	PRE1972	-0.7404	0.282	-2.623	0.009	-1.294	-0.187
264	PRE1984	1.2298	0.360	3.419	0.001	0.525	1.935
265	PRE1988	0.2374	0.247	0.960	0.337	-0.247	0.722
266	PRE1992	-0.1927	0.922	-0.209	0.834	-1.999	1.614
267	PRE1996	2.7137	0.294	9.236	0.000	2.138	3.290
268	PRE2000	-0.7825	0.295	-2.654	0.008	-1.360	-0.205
269	PRE2004	4.4772	0.363	12.350	0.000	3.767	5.188
270	PRE2008	0.8155	0.292	2.792	0.005	0.243	1.388
271	PRE2012	0.7745	0.304	2.545	0.011	0.178	1.371
272	PRE2016	2.0859	0.201	10.378	0.000	1.692	2.480
273	PRE2020	-0.7553	0.284	-2.658	0.008	-1.312	-0.198
274	POST1900	-30.5839	1.423	-21.493	0.000	-33.373	-27.795
275	POST1904	-10.8401	1.512	-7.168	0.000	-13.804	-7.876
276	POST1908	-4.2305	1.698	-2.492	0.013	-7.558	-0.903
277	POST1912	3.1523	1.029	3.065	0.002	1.136	5.168
278	POST1920	8.5944	0.714	12.032	0.000	7.194	9.994
279	POST1924	-0.3393	0.136	-2.491	0.013	-0.606	-0.072
280	POST1928	1.5539	0.351	4.423	0.000	0.865	2.242
281	POST1932	0.2363	0.401	0.589	0.556	-0.550	1.023
282	POST1936	-2.0930	1.145	-1.828	0.068	-4.337	0.151
283	POST1952	-4.1802	0.472	-8.854	0.000	-5.106	-3.255
284	POST1956	-0.1769	0.248	-0.714	0.475	-0.663	0.309
285	POST1960	3.0418	0.240	12.670	0.000	2.571	3.512
286	POST1964	1.9124	0.311	6.155	0.000	1.303	2.521
287	POST1968	2.4417	0.319	7.652	0.000	1.816	3.067
288	POST1972	-0.5846	0.298	-1.962	0.050	-1.169	-0.001
289	POST1988	-0.3048	0.887	-0.344	0.731	-2.043	1.434
290	POST1992	1.8645	0.306	6.100	0.000	1.265	2.464
291	POST1996	0.8210	0.302	2.717	0.007	0.229	1.413
292	POST2000	-1.3470	0.862	-1.563	0.118	-3.036	0.342
293	POST2004	3.2648	0.362	9.030	0.000	2.556	3.973
294	POST2008	-1.5319	0.216	-7.104	0.000	-1.955	-1.109
295	POST2012	6.3429	0.303	20.958	0.000	5.750	6.936
296	POST2016	2.7213	0.279	9.751	0.000	2.174	3.268
297	POST2020	0.8825	0.255	3.458	0.001	0.382	1.383
298	POST2024	1.1720	0.534	2.195	0.028	0.125	2.219

Appendice - risultati OLS 1932

		coef	std err	z	P> z	[0.025	0.975]
150							
151							
152	const	-7.7396	1.358	-5.700	0.000	-10.401	-5.078
153	Avg_Medals_History	0.4971	0.025	19.702	0.000	0.448	0.547
154	GDP	0.2803	0.084	3.341	0.001	0.116	0.445
155	Population	0.5471	0.063	8.644	0.000	0.423	0.671
156	Is_Communist	1.1453	0.177	6.472	0.000	0.798	1.492
157	YEAR1932	0.7210	0.529	1.364	0.173	-0.315	1.757
158	YEAR1936	0.7578	0.472	1.605	0.108	-0.167	1.683
159	YEAR1948	1.6846	0.562	2.997	0.003	0.583	2.786
160	YEAR1952	0.4938	0.425	1.161	0.246	-0.340	1.328
161	YEAR1956	0.1638	0.357	0.458	0.647	-0.536	0.864
162	YEAR1960	0.0837	0.364	0.230	0.818	-0.629	0.796
163	YEAR1964	0.2433	0.400	0.608	0.543	-0.541	1.028
164	YEAR1968	0.1769	0.398	0.444	0.657	-0.604	0.958
165	YEAR1972	0.3196	0.361	0.885	0.376	-0.388	1.028
166	YEAR1976	-0.2504	0.370	-0.677	0.498	-0.975	0.474
167	YEAR1980	0.0889	0.429	0.207	0.836	-0.751	0.929
168	YEAR1984	0.5305	0.385	1.378	0.168	-0.224	1.285
169	YEAR1988	-0.3699	0.336	-1.102	0.271	-1.028	0.288
170	YEAR1992	-0.0364	0.336	-0.108	0.914	-0.695	0.622
171	YEAR1996	0.3657	0.348	1.050	0.294	-0.317	1.049
172	YEAR2000	0.1750	0.349	0.501	0.616	-0.509	0.859
173	YEAR2004	0.0943	0.385	0.245	0.807	-0.661	0.849
174	YEAR2008	-0.0695	0.317	-0.220	0.826	-0.690	0.551
175	YEAR2012	-0.0557	0.343	-0.162	0.871	-0.729	0.617
176	YEAR2016	-0.1595	0.303	-0.526	0.599	-0.754	0.435
177	YEAR2020	-0.1262	0.322	-0.392	0.695	-0.756	0.504
178	OG1932	12.6129	0.665	18.972	0.000	11.310	13.916
179	OG1936	16.4273	0.361	45.554	0.000	15.721	17.134
180	OG1948	-2.6122	0.513	-5.095	0.000	-3.617	-1.607
181	OG1952	1.7425	0.315	5.536	0.000	1.126	2.360
182	OG1956	5.7011	0.201	28.408	0.000	5.308	6.094
183	OG1960	3.8443	0.223	17.221	0.000	3.407	4.282
184	OG1964	2.4786	0.305	8.138	0.000	1.882	3.076
185	OG1968	0.4919	0.290	1.698	0.090	-0.076	1.060
186	OG1976	0.3655	0.262	1.396	0.163	-0.148	0.879
187	OG1984	9.3395	0.471	19.837	0.000	8.417	10.262
188	OG1988	3.1772	0.223	14.258	0.000	2.740	3.614
189	OG1992	1.3325	0.223	5.985	0.000	0.896	1.769
190	OG1996	0.1584	0.437	0.362	0.717	-0.699	1.016
191	OG2000	4.4129	0.226	19.547	0.000	3.970	4.855
192	OG2004	0.2858	0.268	1.068	0.286	-0.239	0.811
193	OG2008	5.3370	0.333	16.044	0.000	4.685	5.989
194	OG2012	2.4077	0.243	9.911	0.000	1.932	2.884
195	OG2016	0.1652	0.195	0.848	0.396	-0.216	0.547
196	OG2020	2.4779	0.217	11.397	0.000	2.052	2.904
197	OG2024	2.0598	0.274	7.517	0.000	1.523	2.597

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198	PRE1932	0.8963	0.448	2.001	0.045	0.018	1.774
199	PRE1936	-1.4850	0.165	-9.013	0.000	-1.808	-1.162
200	PRE1948	0.7218	0.489	1.476	0.140	-0.237	1.680
201	PRE1952	1.0636	0.295	3.606	0.000	0.485	1.642
202	PRE1956	1.8600	0.204	9.130	0.000	1.461	2.259
203	PRE1960	1.3573	0.235	5.766	0.000	0.896	1.819
204	PRE1964	-0.7278	0.283	-2.569	0.010	-1.283	-0.173
205	PRE1972	-0.9761	0.245	-3.977	0.000	-1.457	-0.495
206	PRE1984	1.1129	0.279	3.985	0.000	0.566	1.660
207	PRE1988	-0.0478	0.224	-0.214	0.831	-0.486	0.391
208	PRE1992	-0.5844	0.437	-1.338	0.181	-1.441	0.272
209	PRE1996	2.9532	0.228	12.928	0.000	2.505	3.401
210	PRE2000	-0.0755	0.216	-0.349	0.727	-0.499	0.348
211	PRE2004	1.8192	0.424	4.293	0.000	0.989	2.650
212	PRE2008	0.6605	0.202	3.273	0.001	0.265	1.056
213	PRE2012	-0.0683	0.258	-0.265	0.791	-0.574	0.438
214	PRE2016	1.3933	0.186	7.473	0.000	1.028	1.759
215	PRE2020	-0.9149	0.203	-4.503	0.000	-1.313	-0.517
216	POST1932	0.3689	0.412	0.895	0.371	-0.439	1.177
217	POST1936	-2.1761	0.593	-3.670	0.000	-3.338	-1.014
218	POST1952	-4.4508	0.340	-13.096	0.000	-5.117	-3.785
219	POST1956	0.6597	0.222	2.971	0.003	0.225	1.095
220	POST1960	3.0579	0.216	14.158	0.000	2.635	3.481
221	POST1964	1.3443	0.295	4.554	0.000	0.766	1.923
222	POST1968	1.4576	0.316	4.612	0.000	0.838	2.077
223	POST1972	-1.0897	0.251	-4.336	0.000	-1.582	-0.597
224	POST1988	-0.7018	0.444	-1.580	0.114	-1.572	0.169
225	POST1992	1.6594	0.215	7.726	0.000	1.238	2.080
226	POST1996	0.6237	0.241	2.589	0.010	0.151	1.096
227	POST2000	-1.7297	0.423	-4.089	0.000	-2.559	-0.901
228	POST2004	3.4724	0.282	12.326	0.000	2.920	4.025
229	POST2008	-0.8974	0.187	-4.787	0.000	-1.265	-0.530
230	POST2012	3.6028	0.374	9.627	0.000	2.869	4.336
231	POST2016	2.6031	0.181	14.373	0.000	2.248	2.958
232	POST2020	0.0907	0.223	0.406	0.684	-0.347	0.528
233	POST2024	1.2374	0.281	4.398	0.000	0.686	1.789

Appendice - risultati OLS 1964

	coef	std err	z	P> z	[0.025	0.975]
120						
121	const	-5.7373	1.203	-4.769	0.000	-8.095 -3.379
122	Avg_Medals_History	0.4979	0.033	15.061	0.000	0.433 0.563
123	GDP	0.1059	0.080	1.324	0.185	-0.051 0.263
124	Population	0.5314	0.055	9.713	0.000	0.424 0.639
125	Is Communist	1.0093	0.175	5.781	0.000	0.667 1.352
126	YEAR1964	0.0168	0.391	0.043	0.966	-0.750 0.784
127	YEAR1968	-0.0403	0.386	-0.104	0.917	-0.797 0.717
128	YEAR1972	0.1134	0.352	0.322	0.747	-0.576 0.803
129	YEAR1976	-0.3815	0.366	-1.042	0.297	-1.099 0.336
130	YEAR1980	-0.0183	0.424	-0.043	0.966	-0.848 0.812
131	YEAR1984	0.4250	0.387	1.098	0.272	-0.333 1.183
132	YEAR1988	-0.4786	0.334	-1.435	0.151	-1.132 0.175
133	YEAR1992	-0.1349	0.333	-0.404	0.686	-0.788 0.519
134	YEAR1996	0.2259	0.348	0.650	0.516	-0.456 0.908
135	YEAR2000	0.0675	0.350	0.193	0.847	-0.618 0.753
136	YEAR2004	-0.0245	0.381	-0.064	0.949	-0.771 0.722
137	YEAR2008	-0.0961	0.318	-0.302	0.763	-0.720 0.528
138	YEAR2012	-0.0945	0.349	-0.271	0.787	-0.778 0.589
139	YEAR2016	-0.1581	0.306	-0.516	0.606	-0.758 0.442
140	YEAR2020	-0.1332	0.325	-0.409	0.682	-0.771 0.505
141	OG1964	2.4469	0.297	8.251	0.000	1.866 3.028
142	OG1968	0.3939	0.274	1.435	0.151	-0.144 0.932
143	OG1976	0.4004	0.262	1.527	0.127	-0.113 0.914
144	OG1984	9.4206	0.599	15.720	0.000	8.246 10.595
145	OG1988	3.0613	0.222	13.807	0.000	2.627 3.496
146	OG1992	1.3165	0.225	5.842	0.000	0.875 1.758
147	OG1996	0.3205	0.567	0.565	0.572	-0.791 1.432
148	OG2000	4.4941	0.229	19.644	0.000	4.046 4.942
149	OG2004	0.3026	0.262	1.156	0.248	-0.210 0.815
150	OG2008	5.2707	0.319	16.511	0.000	4.645 5.896
151	OG2012	2.4362	0.266	9.156	0.000	1.915 2.958
152	OG2016	0.0270	0.197	0.137	0.891	-0.359 0.413
153	OG2020	2.5031	0.220	11.383	0.000	2.072 2.934
154	OG2024	2.0719	0.283	7.329	0.000	1.518 2.626
155	PRE1964	-0.8489	0.279	-3.039	0.002	-1.397 -0.301
156	PRE1972	-0.8921	0.238	-3.749	0.000	-1.358 -0.426
157	PRE1984	0.9317	0.283	3.296	0.001	0.378 1.486
158	PRE1988	-0.0896	0.226	-0.397	0.692	-0.532 0.353
159	PRE1992	-0.4749	0.574	-0.828	0.408	-1.599 0.650
160	PRE1996	3.0414	0.232	13.112	0.000	2.587 3.496
161	PRE2000	-0.1023	0.218	-0.470	0.638	-0.529 0.324
162	PRE2004	1.7917	0.414	4.323	0.000	0.979 2.604
163	PRE2008	0.6831	0.221	3.086	0.002	0.249 1.117
164	PRE2012	-0.1727	0.264	-0.654	0.513	-0.690 0.345
165	PRE2016	1.4050	0.186	7.570	0.000	1.041 1.769
166	PRE2020	-0.9019	0.219	-4.110	0.000	-1.332 -0.472
167	POST1964	1.3597	0.296	4.600	0.000	0.780 1.939
168	POST1968	1.4726	0.303	4.858	0.000	0.878 2.067
169	POST1972	-1.1737	0.245	-4.790	0.000	-1.654 -0.693
170	POST1988	-0.5953	0.579	-1.028	0.304	-1.730 0.540
171	POST1992	1.5907	0.216	7.361	0.000	1.167 2.014
172	POST1996	0.6632	0.246	2.694	0.007	0.181 1.146
173	POST2000	-1.5791	0.545	-2.897	0.004	-2.648 -0.511
174	POST2004	3.5855	0.277	12.930	0.000	3.042 4.129
175	POST2008	-0.9444	0.192	-4.926	0.000	-1.320 -0.569
176	POST2012	3.5969	0.368	9.778	0.000	2.876 4.318
177	POST2016	2.6011	0.206	12.640	0.000	2.198 3.004
178	POST2020	-0.0473	0.229	-0.207	0.836	-0.495 0.401
179	POST2024	1.2582	0.279	4.511	0.000	0.712 1.805

Appendice - risultati OLS 1996

		coef	std err	z	P> z	[0.025	0.975]
89							
90							
91	const	-8.1513	1.324	-6.155	0.000	-10.747	-5.556
92	Avg_Medals_History	0.4626	0.029	16.189	0.000	0.407	0.519
93	GDP	0.2678	0.071	3.776	0.000	0.129	0.407
94	Population	0.6314	0.077	8.202	0.000	0.481	0.782
95	Is_Communist	0.6955	0.199	3.495	0.000	0.305	1.086
96	YEAR1996	0.3605	0.313	1.153	0.249	-0.253	0.973
97	YEAR2000	0.1961	0.317	0.619	0.536	-0.425	0.817
98	YEAR2004	0.1640	0.344	0.477	0.633	-0.510	0.838
99	YEAR2008	-0.0085	0.278	-0.031	0.976	-0.554	0.537
100	YEAR2012	0.0417	0.302	0.138	0.890	-0.551	0.634
101	YEAR2016	-0.0592	0.265	-0.224	0.823	-0.578	0.460
102	YEAR2020	-0.0864	0.282	-0.306	0.760	-0.640	0.467
103	OG1996	0.3100	0.462	0.672	0.502	-0.595	1.215
104	OG2000	4.1829	0.223	18.744	0.000	3.746	4.620
105	OG2004	0.0533	0.248	0.215	0.829	-0.432	0.539
106	OG2008	5.1706	0.406	12.731	0.000	4.375	5.967
107	OG2012	2.1406	0.238	9.011	0.000	1.675	2.606
108	OG2016	-0.4140	0.215	-1.922	0.055	-0.836	0.008
109	OG2020	2.0871	0.225	9.280	0.000	1.646	2.528
110	OG2024	1.8567	0.246	7.533	0.000	1.374	2.340
111	PRE1996	2.7471	0.226	12.161	0.000	2.304	3.190
112	PRE2000	-0.2589	0.210	-1.232	0.218	-0.671	0.153
113	PRE2004	1.6282	0.500	3.259	0.001	0.649	2.607
114	PRE2008	0.4347	0.201	2.158	0.031	0.040	0.829
115	PRE2012	-0.6441	0.277	-2.325	0.020	-1.187	-0.101
116	PRE2016	0.9387	0.194	4.849	0.000	0.559	1.318
117	PRE2020	-1.1542	0.199	-5.808	0.000	-1.544	-0.765
118	POST1996	0.2854	0.246	1.159	0.246	-0.197	0.768
119	POST2000	-1.6175	0.427	-3.791	0.000	-2.454	-0.781
120	POST2004	3.1974	0.266	12.005	0.000	2.675	3.719
121	POST2008	-1.1209	0.183	-6.131	0.000	-1.479	-0.763
122	POST2012	3.4247	0.442	7.750	0.000	2.559	4.291
123	POST2016	2.3319	0.185	12.606	0.000	1.969	2.694
124	POST2020	-0.4294	0.247	-1.736	0.083	-0.914	0.055
125	POST2024	0.8916	0.265	3.368	0.001	0.373	1.410

Appendice - Top 40

1900:

USA GER GBR FRA CHN ITA SWE JPN AUS FIN KOR CAN NED BEL ESP GRE BRA
MEX RUS HUN POL ROU TCH BUL UKR SUI DEN CUB BLR YUG NOR KAZ KEN AUT
CZE NZL RSA PRK TUR AZE

1932:

USA GER GBR FRA CHN ITA SWE JPN AUS FIN KOR CAN NED BEL ESP GRE BRA
MEX RUS HUN POL ROU TCH BUL UKR SUI DEN CUB BLR YUG NOR KAZ KEN AUT
CZE NZL RSA PRK TUR AZE

1964:

USA CHN GBR JPN AUS FRA KOR CAN ESP BRA MEX GRE RUS GER HUN ITA ROU
POL BUL TCH CUB YUG NED UKR SWE BLR NZL KEN KAZ DEN FIN CZE SUI PRK
AZE TUR NOR IRI JAM UZB

1996:

USA CHN GBR AUS FRA JPN BRA GRE RUS GER ITA KOR NED UKR CUB CAN HUN
ESP BLR ROU POL NZL KAZ KEN SWE CZE DEN JAM BUL TUR SUI AZE NOR IRI UZB
ETH GEO PRK CRO SRB

Appendice - gruppi CLOSE

- *CLOSE_WEST* (Belgio, Francia, Paesi Bassi)
- *CLOSE_CENTER* (Austria, Repubblica Ceca, Germania, Germania Ovest, Ungheria, Polonia, Slovenia, Slovacchia, Svizzera)
- *CLOSE_GMT1* (*CLOSE_CENTER* + *CLOSE_WEST* + Italia + Spagna): fuso orario GMT+1
- *CLOSE_MAIN* (Francia, Germania, Italia, Spagna): i più forti in Europa continentale
- *CLOSE_WIDE* (*CLOSE_GMT1* + Grecia + UK): gruppo allargato a due ospiti importanti

Le suddivisioni rispettano i confini culturali e spaziali definiti dalla StAGN¹⁰.

VARIABILI



alpha

const



**CLOSE_
<group><year
>**